

From Zero and One to the Electron: Path-Connectedness, the Double Cover, and the Geometric Origin of the Quartic Energy in the 0-Sphere Model

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(Dated: October 10, 2026)

This paper traces where the integers of a model of the electron come from. The model is the 0-Sphere model, in which the electron is two separated kernels exchanging a captured photon. We start from arithmetic: the integers are fixed by 0 and 1, the numbers that do not change when squared are exactly $\{0, 1\}$, and the invertible integers are $\{\pm 1\}$, which is the zero-sphere S^0 . The model's two energy identities share the right-hand side 1, yet the bosonic identity $n_A + n_B = 1$ is solved by a whole segment D^1 , whereas the kernel part of the fermionic identity $(n_A + n_B)^2 = 1$ is solved only by the two endpoints, $S^0 = \partial D^1$; the cross term is exactly the seat that lets energy travel between the two separated points. Path-connectedness therefore enters at the foundations, and the next topological step, a closed path that is not simply connected, yields an integer winding number that the model identifies with electric charge; a point vortex in fluid mechanics gives the picture, and quantized circulation in superfluids shows that the integer and its size come from different sources. We retain and sharpen the earlier results of this paper: the amplitude behind the identity obeys a first-order spinor equation unitarily equivalent to the Dirac rest solutions, and the energy exchange between the two kernels is exactly the interference term between the positive- and negative-energy branches, so that zitterbewegung is read as a real internal oscillation and the negative branch as the absorbing half of a paired emission and absorption rather than as an antiparticle; the complex unit is the symplectic structure of the internal oscillator, and the quartic energies are quadratic forms on the Bloch sphere pulled back through the Hopf double cover. We then define the internal time as the transport cycle between the kernels, fixed by the anomalous magnetic moment through $\beta = 0.04047$, show that it runs exactly β times slower than the Dirac branch phase, and show that the kernel separation is the transport speed times one tick, $\lambda_C = \beta c T_{\text{tr}}$. Metric structure is not needed while the two kernels merely sit; it enters with transport, together with the constitutive relation and the fine-structure constant. We state explicitly that the model derives the integer of charge, its existence, quantization, conservation and sign, but not its size: the fine-structure constant is not derived here, nor is the coupling to the electromagnetic potential or the light-cone structure. The decisive test is experimental: the internal transport speed $\beta c \approx 0.04 c$, fixed with no free parameter, is unmeasured and would confirm or exclude the internal clock.

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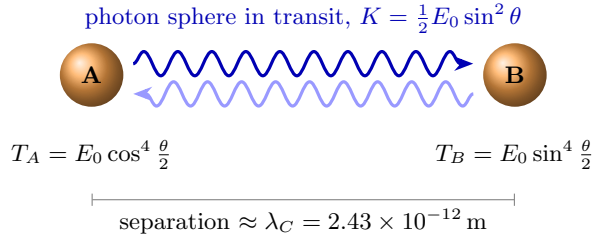


Fig. 1. The electron in the 0-Sphere model

Two kernels A and B , the two points of the zero-sphere S^0 , exchange energy through the photon sphere. The three energies always sum to E_0 , Eq. (I.1).

I. INTRODUCTION

A. Integers in physics and in the model

Physics is full of integers. Charges come in whole multiples, angular momentum comes in half-integer steps [1, 2], and the number of states of a spin- $\frac{1}{2}$ particle is two. In the standard theory these integers are mostly put in by hand: one chooses a symmetry group, and the integers appear as labels of its representations. A model that claims to describe the inside of the electron owes the reader a better answer. It should say where its integers come from, which of them it can derive, and which of them it cannot.

This paper takes up that question for the 0-Sphere model (abbreviated OSM below). Each section states the problem it addresses and why the step is needed, and then gives numbers, a worked example, and a short summary; the paper can be read without the earlier papers of the series. The flow of the whole argument is collected in Fig. 21 at the end.

B. The model in one paragraph

The 0-Sphere model represents the electron not as a point but as a small internal system: two perfect black-body *kernels*, A and B , separated by about one Compton wavelength, together with a captured photon, the *photon sphere*, that carries energy back and forth between them [3]. The name comes from topology: the zero-dimensional sphere S^0 is a pair of points, and the two kernels are such a pair. The bookkeeping of the internal energy is carried by one identity, cited by every paper of the series,

$$1 = \cos^4 \frac{\theta}{2} + \sin^4 \frac{\theta}{2} + \frac{1}{2} \sin^2 \theta, \quad (\text{I.1})$$

multiplied by the total energy E_0 . The first two terms are the thermal potential energies of kernels A and B , and the third is the kinetic energy of the photon sphere in transit. The angle θ advances with time; what clock measures that advance is the subject of Sec. VII.

C. Four results of this paper

Equation (I.1) is an algebraic identity, true for every θ . Its physical content therefore lies not in the fact that it sums to one, but in the structure it hides. We extract four pieces of that structure.

1. *The fourth powers.* Relativistic quantum mechanics expects energies quadratic in the amplitude [4]. Section VI shows that the fourth powers are quadratic energies seen through the spinor double cover.
2. *Boson and fermion.* The photon obeys $\cos^2 + \sin^2 = 1$, the electron the identity above, and both right-hand sides are 1. Section III shows that the difference is the shape of the solution set: a segment versus its two endpoints.
3. *The origin of integers.* Sections II, IV and V trace them from the arithmetic of 0 and 1 through path-connectedness to winding numbers, which the model identifies with charge.
4. *Internal time and distance.* Section VII defines the internal time as the transport cycle, fixed by the anomalous magnetic moment, and builds the kernel separation from it.

Section VIII looks ahead to the comparison of two electrons' clocks and to where the metric enters. Section IX lists what experiment can decide, and Sec. X states the distance from standard theory without softening it. Appendix D derives the topological properties of charge in full.

D. Conventions: three layers and two kinds of statement

Two habits will prevent most misreadings.

a. Three layers. The two kernels are described at three levels, and we never mix them. The *amplitude* layer carries $a = \cos(\theta/2)$ and $b = \sin(\theta/2)$, which can be negative. The *occupation* layer carries $n_A = a^2$ and $n_B = b^2$, which are non-negative and sum to one; the kernel *temperature* lives here. The *energy* layer carries $T_A = E_0 n_A^2$ and $T_B = E_0 n_B^2$. The fourth power in Eq. (I.1) is therefore a fourth power of the amplitude, but only a square of the occupation.

b. Two kinds of statement. Following the structural overview of the series [5], we mark statements that belong to textbook physics or mathematics as **[standard]** and statements that belong to the interpretation of the 0-Sphere model as **[OSM reading]** wherever confusion could arise. A reader may accept every **[standard]** statement and still reject the model; the paper is written so that this division is always visible.

E. What goes in and what comes out

a. Why this is needed. A reader returning to this paper after some time should be able to see at once which statements are assumed and which are derived. Table I lists the inputs of the argument and the outputs, each with the section where it appears.

F. Organization of the paper

The argument runs along one main line, and every step of it is in the body text. Three kinds of aid surround the main line, each visibly marked. *Worked examples and tables* put numbers on each claim, and each number can be reproduced by hand. Boxes headed *A common misreading* state an objection likely to arise at that point and answer it. Boxes headed *What this section established* close each section with the result in two or three sentences. Table II collects every symbol with the layer it lives on; Fig. 21 at the end shows the whole route on one page; and Appendix A maps each numerical check to the claim it supports.

II. ZERO AND ONE: THE ARITHMETIC SEED

a. Motivation. The paper begins with the arithmetic of 0 and 1 because the three facts below are the patterns that the rest of the paper finds again inside the electron. Seeing them first in their simplest setting makes the later sections recognisable rather than surprising.

A. The integers are fixed by 0 and 1

[standard] As an additive group, the integers $\mathbb{Z}$ are generated by 1 alone: every integer is $1 + 1 + \dots$ or its negative, and 0 is the empty sum. As a ring, $\mathbb{Z}$ is fixed by 0 and 1 together [7]: for any ring R there is exactly one structure-preserving map $\mathbb{Z} \rightarrow R$, the one sending $0 \mapsto 0$ and $1 \mapsto 1$. In this precise sense, once 0 and 1 are given, the whole of $\mathbb{Z}$ is given. This is prior to any notation for numbers: a positional system with base b needs the digits $0, \dots, b-1$, so the decimal system needs ten digits, and only base two gets by with 0 and 1. The structure, not the notation, is what we use.

B. The numbers that do not change when squared

[standard] A number is *idempotent* if squaring it changes nothing, $n^2 = n$. Among the real numbers the solutions are exactly

$$n(n-1) = 0 \iff n \in \{0, 1\} = \mathbb{Z} \cap [0, 1]. \quad (\text{II.1})$$

Between 0 and 1 every number shrinks when squared; only the two ends survive. Section III meets this equation again, as the condition under which the electron's kernel energies alone add up to the total.

C. The invertible integers form the zero-sphere

[standard] Only two integers have integer inverses, $+1$ and -1 . They form the group of units

$$\mathbb{Z}^\times = \{+1, -1\} = S^0, \quad (\text{II.2})$$

the zero-dimensional sphere: the set of points at unit distance from the origin of the real line. [OSM reading] The same two-element set appears twice inside the electron: as the sign flip $\chi \rightarrow -\chi$ of the amplitude after one full turn (Sec. VI), and as the two signs of charge, $e^\mp$ (Sec. V).

D. Where the standard theory gets its integers

a. Why this is needed. To judge what the 0-Sphere model adds, the reader should see how the established theory produces integers. Table III lists four textbook examples. In every one, the integer is *topological or algebraic* and the *size* of the quantum is a separate input; this is the pattern the model will repeat in Sec. V.

What this section established. The integers are fixed by 0 and 1. The numbers unchanged by squaring are exactly 0 and 1. The invertible integers are ± 1 , which is the zero-sphere S^0 . In established physics, integers come from symmetry or topology while the size of each quantum is a separate input.

III. ONE NORMALIZATION, TWO PICTURES

a. Motivation. The photon's identity and the electron's identity both equal one. The difference between the two particles therefore lies not in the total but in *who carries it*.

A. The same one, read at two degrees

Write the occupations of the two kernels as $n_A = \cos^2(\theta/2)$ and $n_B = \sin^2(\theta/2)$. The bosonic identity of the model [14] and the fermionic identity (I.1) are then

$$\text{boson: } n_A + n_B = 1, \quad (\text{III.1})$$

$$\text{fermion: } (n_A + n_B)^2 = n_A^2 + n_B^2 + 2n_A n_B = 1. \quad (\text{III.2})$$

The second is the square of the first, and $2n_A n_B = \frac{1}{2} \sin^2 \theta$ is the cross term of Eq. (I.1). The same normalization is read at degree one and at degree two [14]. Expanded, the

Table I. Inputs and outputs of the argument

Kind	Statement	Status	Where
Input	Two kernels and a photon sphere; the energy identity (I.1)	Model postulate [3]	Sec. I
Input	Simple-harmonic motion of the photon sphere; energy gradient between the kernels	Model postulate [3, Secs. IV, VI 6]	
Input	Anomalous magnetic moment read as the contraction of the half-turn	Model reading [6]; Sec. VII a_e measured	
Output	Boson = D^1 , fermion kernels = $S^0 = \partial D^1$; degrees 1 and 2	Derived	Sec. III
Output	Charge as a protected winding number: existence, integer, conservation, bound, sign	Derived	Sec. V, App. D
Output	Rest-frame equivalence with the Dirac solutions; quartic = Hopf pullback	Derived	Sec. VI
Output	Internal time $\beta mc^2/\hbar$; two clocks a factor β apart; $\lambda_C = \beta c T_{tr}$	Derived (given the inputs)	Sec. VII
Not derived	Size of charge (α); coupling to A_μ ; light cone; spin sector	Open	Secs. VIII, X

“Derived” means derived in this paper from the inputs above it, with the derivation checked numerically (Appendix A). The inputs are the model’s assumptions; the paper does not argue for them beyond citing where they were introduced.

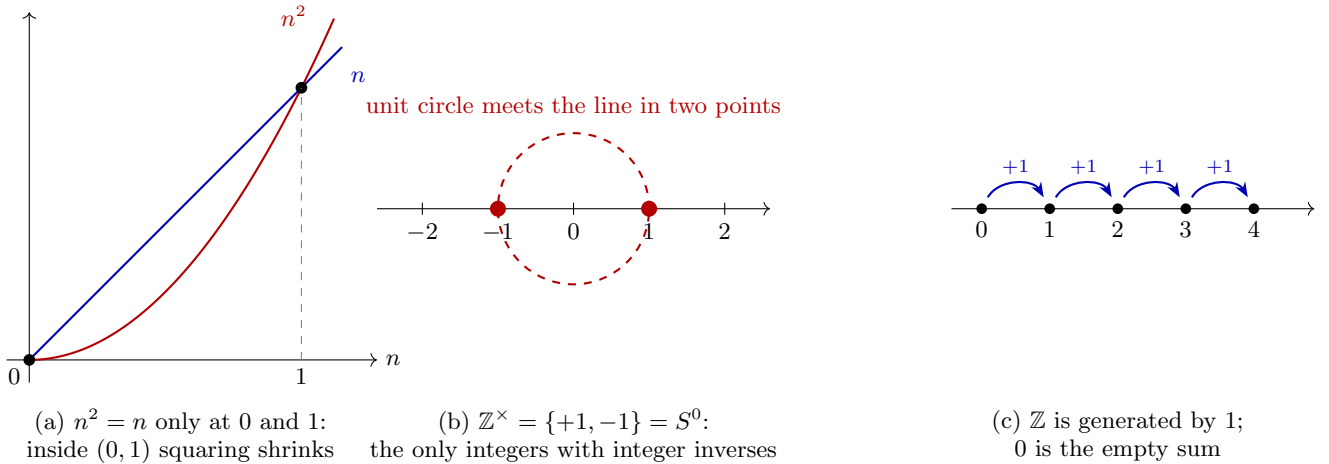


Fig. 2. Three facts about 0 and 1 that reappear inside the electron

(a) The idempotent equation $n^2 = n$ is solved only by 0 and 1; it returns in Eq. (III.3) as the condition for the kernels alone to carry the whole energy. (b) The invertible integers form the zero-sphere, the two points at unit distance from the origin of the line; they return as the sign of the amplitude and as the sign of charge. (c) Once 0 and 1 are given, every integer is given.

fermionic identity has three terms: one seat for kernel A , one for the photon sphere in transit, and one for kernel B .

B. A worked example: one full turn, step by step

a. *Why this is needed.* Equation (I.1) holds for every θ , and a reader may suspect that an identity true for all values says nothing. Following one turn in steps of $\pi/2$ shows what the identity *records*: which seat holds the

Table II. Symbols used in this paper and the layer each lives on

Symbol	Meaning	Layer	First appears
θ	Internal angle; advances with internal — time		Eq. (I.1)
a, b	Amplitudes of kernels A, B : $\cos(\theta/2), \sin(\theta/2)$	Amplitude (can be negative)	Sec. ID
$\chi = (a, b)^\top, z = a + ib$	Kernel rotor; the same pair as a point in the plane	Amplitude	Secs. VI, V
n_A, n_B	Occupations a^2, b^2 ; shares of the total; $n_A + n_B = 1$	Occupation (temperature here)	Sec. III lives
T_A, T_B, K	Kernel energies $E_0 n_A^2, E_0 n_B^2$; transit energy $2E_0 n_A n_B$	Energy	Eq. (I.1)
$\mathbf{s} = (s_x, s_y, s_z)$	Bloch vector $\chi^\dagger \boldsymbol{\sigma} \chi = (\sin \theta, 0, \cos \theta)$	Occupation (full angle)	Sec. VI
D^1, S^0	The segment $[0, 1]$; its two endpoints $\{0, 1\}$	Occupation	Prop. III.1
W, Q	Winding number of z around the origin; Wronskian charge	Amplitude	Eq. (V.2)
β	0.04047; transport speed over c , fixed by a_e	Transport	Eq. (VII.2)
$T_{\text{tr}}, \nu_{\text{tr}}$	One tick of internal time and its frequency	Transport	Table VIII
$\mathcal{T}_{A \rightarrow B}, \mathcal{T}_{(AB) \rightarrow (A'B')}$	One tick inside an electron; comparison between two electrons	Transport / electrons	between Eqs. (VII.3), (VIII.4)
λ_C	Compton wavelength h/mc ; the kernel separation	Distance	Eq. (VII.4)

The three layers of Sec. ID are amplitude, occupation and energy; transport and distance are added in Sec. VII. A symbol is never moved from its layer: in particular temperature is an occupation-layer quantity, and the winding number is an amplitude-layer quantity that is invisible on the occupation layer.

energy at each moment, and how the amplitude changes sign while every observable returns. Table IV gives the numbers.

Three features of the table are used later. The kernels alone carry the whole energy only in the rows where one occupation is 0 and the other is 1 (Proposition III.1). The transit energy is largest exactly when the two occupations are equal. And the amplitude columns change sign after one turn while nothing observable does; the integer of Sec. V will live in those amplitude columns.

A common misreading. “Equation (I.1) is just $(\cos^2 + \sin^2)^2 = 1$; an identity that holds for every angle cannot carry physics.”

The identity itself carries no physics; its *decomposition* does. Which three seats the unit is split into, and how the split changes along the cycle, is the content, exactly as the identity $\cos^2 + \sin^2 = 1$ carries no physics while the split between kinetic and potential energy of a pendulum does. The model’s claim is that the three terms are three physical seats, and the rest of this paper draws consequences from that claim.

Table III. Four integers of established physics and where each comes from

Quantity	The integer	Its origin	The size of one unit
Angular momentum	$j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$	Irreducible representations of $\hbar$ the rotation group $SU(2)$ [2]	
Dirac's magnetic monopole	$eg = n \hbar c/2$ (Gaussian units)	Single-valuedness of the wave function around the string [8]	Fixed by e and $\hbar$
Superconducting flux	$\Phi = n \hbar/2e$	Winding of the condensate phase around the ring [9–11]	$\hbar/2e = 2.068 \times 10^{-15}$ Wb
Quantum Hall conductance	$\sigma_{xy} = \nu e^2/h, \nu \in \mathbb{Z}$	A Chern number of the occupied bands [12, 13]	e^2/h

In each row the integer is fixed by symmetry or topology and is exact; the unit it multiplies is set by constants measured independently. None of these theories derives the fine-structure constant. The 0-Sphere model will be held to the same division: it derives an integer, and the size of the unit is a separate question.

Table IV. One full turn of the internal angle, in steps of $\pi/2$

θ	a	b	n_A	n_B	T_A/E_0	T_B/E_0	K/E_0	Sum	s_z	s_x
0	+1	0	1	0	1	0	0	1	+1	0
$\pi/2$	+0.707	+0.707	0.5	0.5	0.25	0.25	0.5	1	0	+1
π	0	+1	0	1	0	1	0	1	-1	0
$3\pi/2$	-0.707	+0.707	0.5	0.5	0.25	0.25	0.5	1	0	-1
2π	-1	0	1	0	1	0	0	1	+1	0
$5\pi/2$	-0.707	-0.707	0.5	0.5	0.25	0.25	0.5	1	0	+1
3π	0	-1	0	1	0	1	0	1	-1	0
$7\pi/2$	+0.707	-0.707	0.5	0.5	0.25	0.25	0.5	1	0	-1
4π	+1	0	1	0	1	0	0	1	+1	0

Read the table in three ways. Down the energy columns: the energy moves from kernel A ($\theta = 0$) through the transit seat ($\theta = \pi/2$, where the photon sphere carries half of everything) to kernel B ($\theta = \pi$), and back. Across each row: the three energies always sum to one. Down the amplitude columns: at $\theta = 2\pi$ every occupation, energy and Bloch component has returned, but $(a, b) = (-1, 0)$; the amplitude returns to $(+1, 0)$ only at $\theta = 4\pi$. This is the double cover, visible in plain numbers.

C. Where the kernels alone carry the whole

Table V gives, at five points along the cycle, how much of the total the two kernels hold by themselves, without the photon sphere.

The pattern is exact. On the segment $n_A + n_B = 1$, the condition that the kernel terms alone give one is

$$\begin{aligned} n_A^2 + (1 - n_A)^2 = 1 &\iff n_A(n_A - 1) = 0 \\ &\iff n_A \in \{0, 1\}, \end{aligned} \quad (\text{III.3})$$

which is the idempotent equation (II.1). The kernels can carry the whole where the occupation does not change when squared.

D. Two solution sets: a segment and its two ends

We can now state the difference between the two particles in the language of shapes.

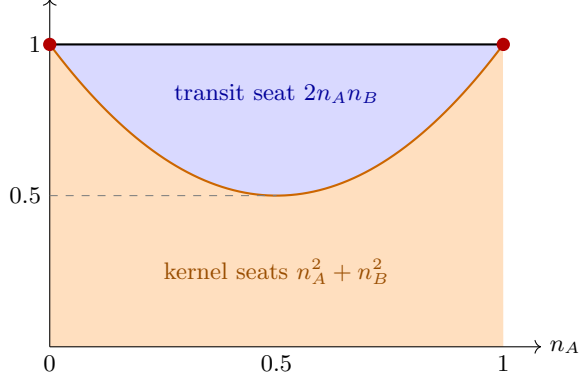
Proposition III.1 (Two solution sets). *On the occupation segment $\{(n_A, n_B) : n_A, n_B \geq 0, n_A + n_B = 1\}$, the bosonic identity holds on the whole segment $D^1 = [0, 1]$, which is path-connected and one-dimensional. For every degree $p > 1$, the kernel part $n_A^p + n_B^p = 1$ holds only at the two endpoints $\{0, 1\} = S^0 = \partial D^1$, which are not path-connected.*

Proof. For $p > 1$ the function $f(x) = x^p + (1 - x)^p$ is strictly convex on $[0, 1]$ with $f(0) = f(1) = 1$, so $f < 1$ in the interior. For $p = 1$, $f \equiv 1$. $\square$

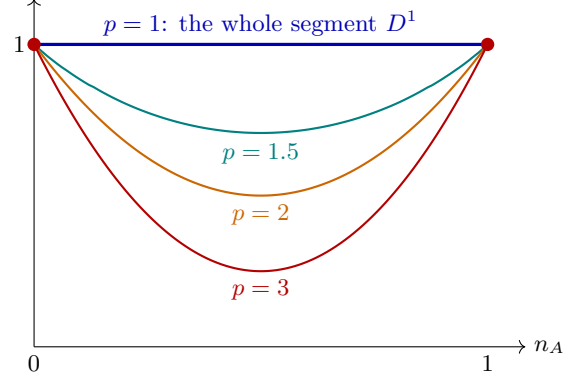
Table V. How the unit total is shared between the kernel seats and the transit seat

Quantity	$n_A = 0$	$n_A = 0.25$	$n_A = 0.5$	$n_A = 0.75$	$n_A = 1$
Kernel seats $n_A^2 + n_B^2$	1	0.625	0.5	0.625	1
Transit seat $2n_A n_B$	0	0.375	0.5	0.375	0
Total $(n_A + n_B)^2$	1	1	1	1	1

The kernels alone hold the whole only at the two ends of the cycle. Everywhere in between they fall short, and the shortfall is exactly the energy of the photon sphere in transit. The minimum 0.5 at $n_A = 0.5$ is the floor $E_0/2$ discussed in Ref. [15].



(a) the total is always 1; the kernels alone reach it only at the two ends



(b) $n_A^p + n_B^p = 1$ holds on all of D^1 for $p = 1$, but only on $S^0 = \{0, 1\}$ for every $p > 1$

Fig. 3. One normalization, two solution sets

(a) Along the cycle, the fermionic total $(n_A + n_B)^2 = 1$ is shared between the kernel seats (orange) and the transit seat (blue); compare Table V. (b) Proposition III.1: for the boson ($p = 1$) every point of the segment is a solution; for any $p > 1$ the kernel terms reach one only at the two endpoints (red dots). The boson is the segment; the fermion's kernels are its boundary.

[OSM reading] The boson is the segment; the fermion's kernels are its boundary. *Zero is the boundary of one:* $S^0 = \partial D^1$. The cross term $2n_A n_B$ is what fills the segment back in, so that the fermionic total is again one at every point. It is the seat that energy occupies while travelling between two points that are otherwise disconnected. The same two points appear in two further places in the corpus: as the two fold points when the amplitude circle is folded onto the occupation segment, which are the critical points of the internal Morse function [16], and as the two branch points $\pm mc^2$ of the square root in the Dirac energy, where they appear as the edges of the Joukowski plate of the trunk analysis [17].

E. Choice of degree: two independent criteria

Proposition III.1 separates $p = 1$ from every $p > 1$; it does not by itself pick $p = 2$. Two independent criteria do, and they meet at the same answer.

a. Harmonicity. The difference of kernel energies $D_p = n_A^p - n_B^p$ should be a pure harmonic oscillation, $D_p \propto \cos \theta$, so that the internal motion is simple-harmonic as assumed since Ref. [6]. Writing $u = \cos \theta$, one has $D_p = [(1+u)^p - (1-u)^p]/2^p$, and the cubic term vanishes only if $\binom{p}{3} = p(p-1)(p-2)/6 = 0$. The real solutions are $\{0, 1, 2\}$, and $p = 0$ is degenerate. Harmonicity selects $\{1, 2\}$ from the whole real line, without assuming that p is an integer.

b. A worked check of harmonicity. For $p = 2$ the difference factorizes, $D_2 = n_A^2 - n_B^2 = (n_A - n_B)(n_A + n_B) = \cos \theta \cdot 1$, a pure harmonic. For $p = 3$ it does not: $D_3 = n_A^3 - n_B^3 = (n_A - n_B)(n_A^2 + n_A n_B + n_B^2) = \cos \theta (1 - \frac{1}{4} \sin^2 \theta)$, which contains $\cos^3 \theta$ and is no longer simple-harmonic. For $p = 1$, $D_1 = \cos \theta$ trivially. The factor $(n_A + n_B)^{p-1}$ that would be needed to make D_p harmonic is available only at $p = 1$ and $p = 2$.

c. Counting seats. The expansion of $(n_A + n_B)^p$ has $p + 1$ terms. Two kernels and one carrier in transit need three seats, hence $p = 2$. A photon has no kernels and is itself the carrier, so it needs only the two seats of

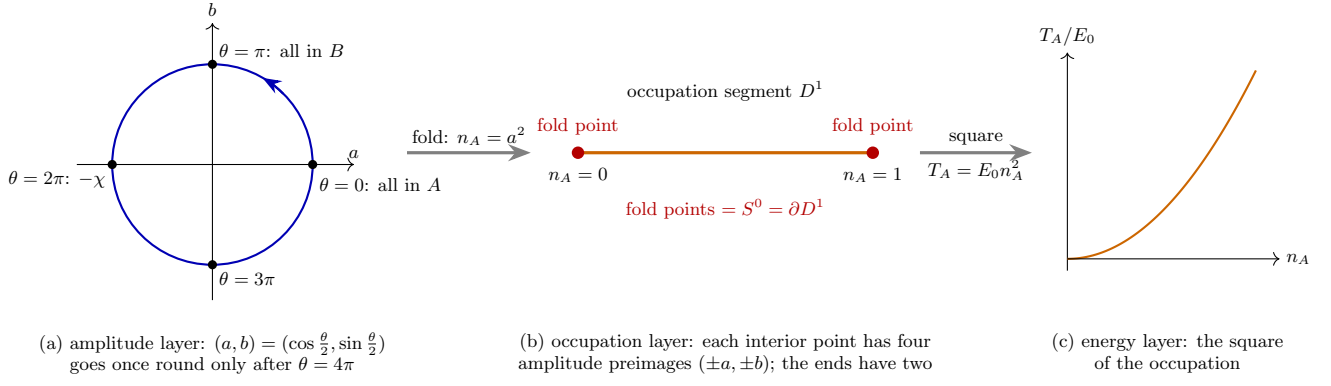


Fig. 4. Three layers: amplitude, occupation, energy

(a) The amplitude can be negative and carries the sign flip $\chi \rightarrow -\chi$ after one full turn of θ . (b) Squaring folds the circle onto the occupation segment; the two fold points are the zero-sphere, the boundary of the segment, and the critical points of the internal oscillation. Temperature lives on this layer. (c) The kernel energy is the square of the occupation, hence the fourth power of the amplitude. The winding number of Sec. V lives on layer (a), where the curve can go round the origin; it is invisible on the folded layer (b).

$p = 1$. Counting seats picks one member of the pair that harmonicity allowed.

F. What does *not* separate them

A caution is needed, because an earlier draft of this paper stated otherwise. One might expect the spinor double cover to be what separates fermion from boson. It is not. The sign flip $a \rightarrow -a$, $b \rightarrow -b$ under $\theta \rightarrow \theta + 2\pi$ follows from the definition $a = \cos(\theta/2)$, $b = \sin(\theta/2)$ alone, and holds for every degree p ; the number of critical points of the internal oscillation is also independent of p . The double cover is shared ground, not the dividing wall. What divides the two pictures is Proposition III.1 together with the seat count: whether the energy can be split into kernel seats globally, at every point of the cycle. We also do not claim that this division is the same as the spin–statistics division of standard theory, whose origin is the topology of exchanging identical particles in three dimensions [18–20]; the 0-Sphere model reaches the same two-way choice at a different place and for a different reason.

What this section established. The boson’s identity is solved by the whole occupation segment D^1 ; the kernel part of the fermion’s identity only by its two endpoints $S^0 = \partial D^1$. The cross term is the seat of energy in transit. Harmonicity selects degrees 1 and 2, the seat count selects 2 for a two-kernel particle, and the double cover is shared by both pictures rather than separating them.

IV. PATH-CONNECTEDNESS: JOINING TWO SEPARATED POINTS

a. Motivation. The fermion’s kernels are two disconnected points, and a disconnected set has no paths; energy

nevertheless travels between them. The resolution is that the two points sit inside a larger connected region, and the cross term is the bookkeeping of the journey through it.

A. From two points to a connected domain

[standard] A space is *path-connected* if any two of its points can be joined by a continuous path inside it. The zero-sphere S^0 is the simplest space that is not: its two points have nothing between them. [OSM reading] Reference [21] resolves this by embedding: the two kernels are discrete locations in a path-connected compact domain $\mathcal{D}$, so that discrete localisation and continuous transport coexist. Reference [22] turns this into a chain of statements: S^0 is not path-connected; it embeds in a sphere; the Bonnet–Myers bound limits the diameter [23]; the domain is path-connected and the two kernels are not antipodal; there is a unique thermal geodesic joining them; and the two-point phase functional along that geodesic is well defined. Proposition III.1 adds the energy-side counterpart: the transit seat $2n_A n_B$ is exactly what the energy budget needs for that journey.

B. Gradients, curls, and the first layer of the Helmholtz decomposition

a. Why this is needed. The energy that drives the photon sphere is a difference of two scalar quantities, the kernel energies. A vector field that is the gradient of a scalar has no curl. A reader who knows electromagnetism will immediately ask where curl, and with it the magnetic field, could ever come from. Fluid mechanics answers this cleanly.

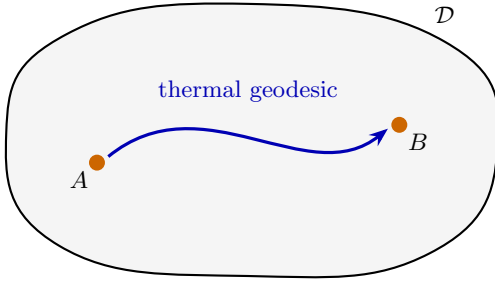


Fig. 5. Two separated points inside a connected domain

The kernels A and B form S^0 , which on its own has no path between its points. Embedded in the path-connected domain $\mathcal{D}$ of Refs. [21, 22], they are joined by a unique thermal geodesic along which the photon sphere travels.

[standard] The Helmholtz decomposition writes a velocity field as

$$\mathbf{u} = \nabla\phi + \nabla \times \boldsymbol{\psi} + \mathbf{h}, \quad (\text{IV.1})$$

a gradient part, a rotational part, and a harmonic part [24, 25]. The harmonic part $\mathbf{h}$ has no curl and no divergence, yet is not a gradient; it exists only when the domain has holes. In differential-form language this is the Hodge decomposition [26], and the harmonic part counts the holes.

[OSM reading] The first layer of the electron is the gradient part. The drive is the gradient of the energy difference $D = (T_A - T_B)/E_0 = \cos\theta$, and the kinetic energy of the photon sphere is $\frac{1}{2}(dD/d\theta)^2 = \frac{1}{2}\sin^2\theta$, the cross term of Eq. (I.1). On a segment, every curl-free flow is a gradient, so at this layer nothing circulates. Two cautions apply. First, D is a function of the internal angle θ , not yet of position in space; turning an internal gradient into a spatial one requires distance, which enters only in Sec. VII. Second, the other two parts of Eq. (IV.1) do appear, but at different layers: the harmonic part in Sec. V, the rotational part in Sec. VIII.

A common misreading. “If the two kernels are disconnected points, the photon sphere must jump between them.”

No jump is implied. The kernels are discrete locations inside a connected domain (Refs. [21, 22]); the photon sphere travels a continuous path, the thermal geodesic. What is discrete is where energy can *rest*; what is continuous is the path along which it *moves*. The energy budget records both: the kernel seats for resting, the transit seat for moving.

What this section established. Path-connectedness is the property the two kernels lack and the transport supplies. The drive of the photon sphere is the gradient of the energy difference, the first layer of the Helmholtz decomposition; at this layer nothing circulates.

V. CLOSING THE PATH: WINDING NUMBERS AND CHARGE

a. Motivation. So far there is a segment and its two endpoints. An integer that is conserved and carries a sign, as charge does, cannot live on a segment: a segment has no holes, so nothing can wind around anything. An integer appears when the path closes into a loop around a point it cannot cross.

A. From the segment to the circle

[standard] Two elementary operations relate the segment and the circle. Folding a circle onto a diameter gives a segment, with the two fold points forming S^0 ; conversely, gluing the two endpoints of a segment gives a circle, $D^1/S^0 \cong S^1$. The circle is path-connected but not simply connected: a loop that goes once around cannot be shrunk to a point, and the number of times a loop goes around is an integer, $\pi_1(S^1) \cong \mathbb{Z}$ [27, 28].

[OSM reading] The occupation segment is the shadow of the amplitude circle $z = a + ib$, folded at the two kernel points. The integer lives on the circle, not on its shadow. The circle encloses the origin of the amplitude plane only if the two kernels never empty simultaneously, and it closes on itself only if the two kernels are *locked* to the same frequency, which is the normalization $|z| = 1$ [3, 29]. Both conditions together give an integer that cannot change continuously: the winding number of z around the origin. That a quarter-period stagger between the two kernels turns their scalar oscillation into a rotation was shown in Ref. [30].

a. A worked example: why lock and normalization go together. Keep the two kernels at the same frequency but let the second lag by a phase δ : $a = \cos t$, $b = \sin(t + \delta)$. Figure 7 draws the curve for four values of δ . A short calculation (Appendix D) gives the closest approach to the origin in closed form,

$$\min_t |z| = \sqrt{1 - |\sin\delta|}, \quad (\text{V.1})$$

which is 1, 0.707, 0.366 and 0 for $\delta = 0, \pi/6, \pi/3, \pi/2$. The distance from the origin is the *gap* that protects the winding number: as long as it is positive, no small disturbance can change the integer. Only at $\delta = 0$ and $\delta = \pi$ is the gap the full radius at every instant, and only there is the normalization $n_A + n_B = 1$ satisfied at every instant. At $\delta = \pi/2$ the gap closes, the curve passes through the origin, and the integer is lost.

B. A worked example from fluid mechanics: the point vortex

[standard] A point vortex in a plane [31, 32] has the velocity field $u_\theta = \Gamma/(2\pi r)$. Its vorticity vanishes at every

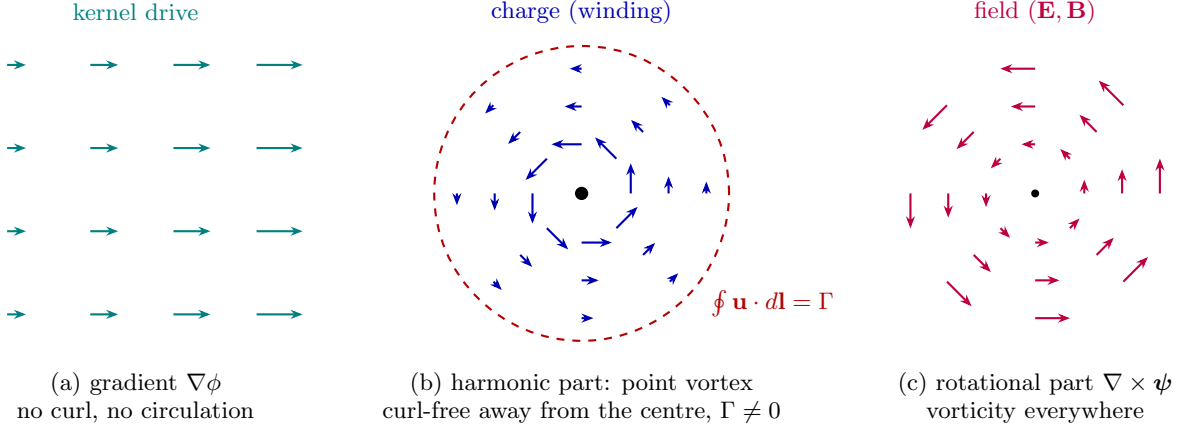


Fig. 6. The Helmholtz decomposition and the three layers of the electron

(a) A gradient flow has no curl and no circulation; this is the drive of the photon sphere by the kernel energy difference (Sec. IV). (b) A point vortex, $u_\theta = \Gamma/(2\pi r)$, is curl-free at every point except the centre, yet carries circulation Γ around any loop that encloses the centre; this is the picture of charge as a winding number (Sec. V). (c) Rigid rotation, $u_\theta \propto r$, has vorticity everywhere; this is the layer at which clock comparisons among several electrons can carry curvature (Sec. VIII).

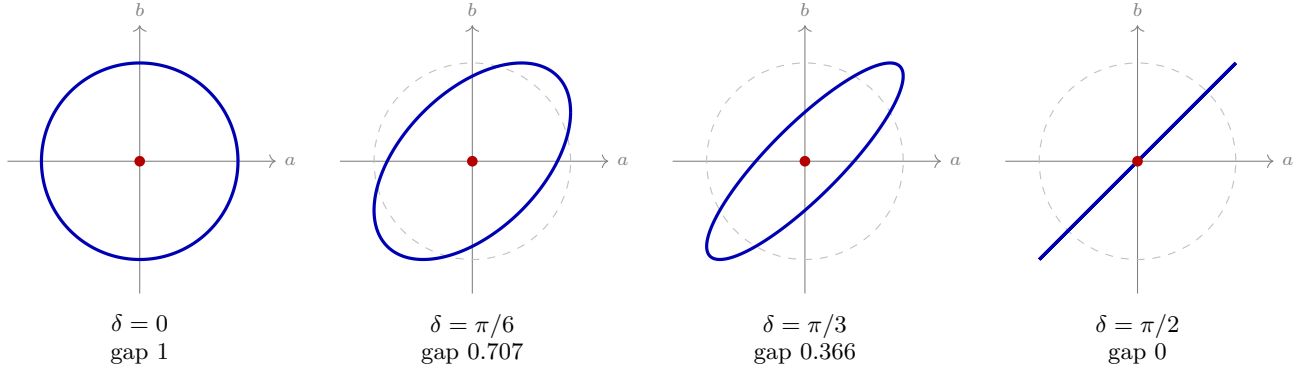


Fig. 7. Closing the gap: from a locked circle to a segment through the origin

The curve $z = \cos t + i \sin(t + \delta)$ for four relative phases. The dashed circle is the normalized orbit. As δ grows, the circle flattens into an ellipse that comes closer to the origin, with closest approach $\sqrt{1 - |\sin \delta|}$, Eq. (V.1); at $\delta = \pi/2$ it collapses to a segment through the origin and the winding number ceases to exist.

point except the origin, yet the circulation around any loop enclosing the origin is $\oint \mathbf{u} \cdot d\mathbf{l} = \Gamma \neq 0$. The velocity potential $\phi = \Gamma\varphi/(2\pi)$, with φ the polar angle, exists locally but changes by Γ on each turn, so it is not a single-valued gradient. This is the harmonic part $\mathbf{h}$ of Eq. (IV.1) in its simplest form. Kelvin's theorem says that the circulation is conserved by ideal flow [33]. Curl-free, not a gradient, conserved, and carrying a sign set by the sense of rotation: these are the four properties we want for charge.

C. Charge as a protected winding number

[OSM reading] Following the trunk analysis of the corpus (trunk T2, Sec. T2.4–T2.5) [17], the winding number of the locked two-kernel amplitude is identified

with electric charge. Its content can be stated in one quantity, the Wronskian of the two amplitudes,

$$Q = \frac{2}{\omega} (a \dot{b} - b \dot{a}), \quad (\text{V.2})$$

which equals +1 for the rotor $a = \cos(\omega t/2)$, $b = \sin(\omega t/2)$ and −1 for the reversed sense (Fig. 8a,b). Appendix D derives each statement below. Each property of charge then has a standard mathematical source, as Table VI shows. For a detuned pair, with the two kernels at different frequencies, the closed form of the winding number is the mod-4 Dirichlet character [34], $W = \chi_4(q)$ for odd frequency ratios $p : q$, which takes only the values $0, \pm 1$, that is, the values $\{0\} \cup S^0$; this closed form has been checked against direct counting for all coprime odd pairs up to 301. A detuned pair has no protected integer and is neutral, which the model reads as the neutrino. Earlier papers of the series attributed charge to a single

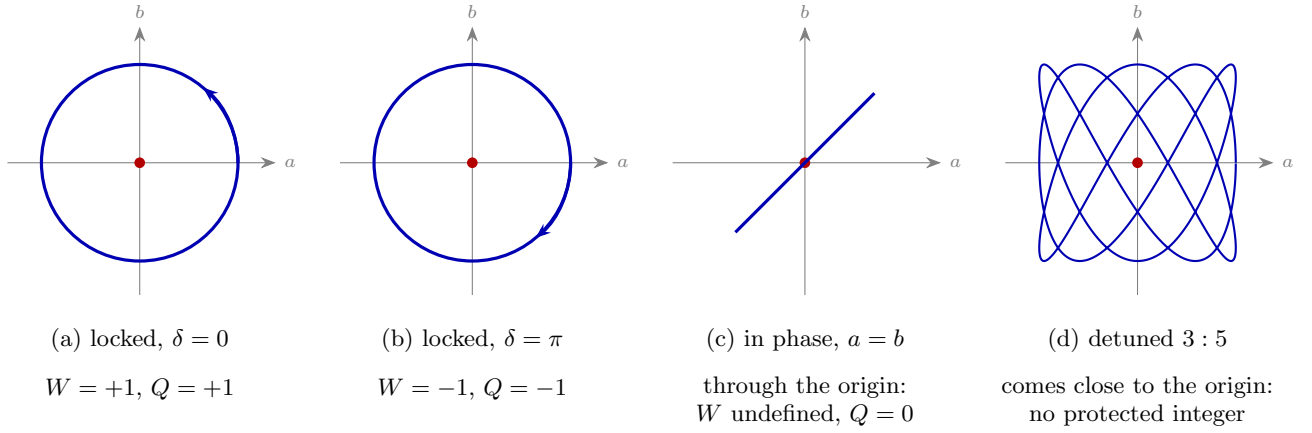


Fig. 8. Charge as a winding number in the amplitude plane

The point $z = a + ib$ records the two kernel amplitudes. (a), (b) When the kernels are locked and normalized, z stays on the unit circle and goes round the origin; the two senses of rotation are the two signs of charge. (c) If the two amplitudes oscillate in phase, the curve collapses to a segment through the origin and no winding can be defined; there is also no energy difference and no zitterbewegung. (d) If the kernels run at different frequencies, here 3 : 5, the curve is a Lissajous figure that approaches the origin; nothing protects an integer, and the system is neutral. Appendix D gives the proofs.

term of the identity [35], discussed its localization [36], and asked whether the symmetry of the kernel pair is $SO(2)$ or $O(2)$ [37]. The winding-number reading corrects the first, since a single term oscillates and is not conserved, and answers the last: the reflection $b \rightarrow -b$ that reverses the winding lies outside $SO(2)$ and is charge conjugation.

D. Worked examples: winding numbers of detuned pairs

a. Why this is needed. The claim that detuned kernels carry no protected integer is easy to state and hard to picture. Table VII lists the winding numbers of the curves $z(\tau) = \cos p\tau + i \sin q\tau$ for small frequency ratios, computed by counting turns around the origin, and Fig. 9 draws three of them; such curves are the figures of Lissajous [39].

E. The integer and its size come from different places

a. Why this matters. Table VI ends with two rows marked *not derived*. A reader may take this as a defect. Quantized circulation in condensed matter shows that it is the normal structure of any quantized winding.

[**standard**] In superfluid helium-4 the circulation around a vortex is $\Gamma = n\kappa$ [40–42]: the integer n is the winding of the condensate phase, while the size of one quantum, $\kappa = h/m_{\text{He}} = 9.97 \times 10^{-8} \text{ m}^2/\text{s}$, comes from Planck’s constant and the atomic mass. In a superconducting ring the magnetic flux is $\Phi = n h/(2e)$ [10, 11] with $h/(2e) = 2.068 \times 10^{-15} \text{ Wb}$: again the integer is a winding and the quantum is a separate

input. [**OSM reading**] Charge has the same structure: the integer is the winding number of Table VI, and the size of one quantum is a separate input that the model does not yet supply.

It also matters which number is to be derived. Since 2019 the elementary charge e is a defined constant of the SI [43]. The physical, unit-free number is the fine-structure constant

$$\alpha = \frac{e^2}{4\pi\epsilon_0\hbar c} = \frac{1}{137.036}. \quad (\text{V.3})$$

The numerical values of the constants used in this paper are those of CODATA 2018 [44]. In a connection $D\theta = d\theta + qA$, the charge q plays the role of a conversion rate between a dimensionless internal phase and the electromagnetic potential, in the same way that β converts one internal rate into another (Sec. VII). The open problem is therefore not “the mechanism of charge” in general, but one number: where α comes from in the model.

Statement of scope. The 0-Sphere model, as developed in this paper, derives the *integer* of electric charge: its existence, its quantization, its conservation, its bound $|Q| \leq 1$, and its sign. It does *not* derive the *size* of the charge. The fine-structure constant $\alpha = 1/137.036$ is not derived here, and neither is the coupling of the winding number to the electromagnetic potential.

Table VI. What the model derives about charge, and what it does not

Property of charge	Source in the model	Mathematical tool	Status
Existence	Two kernels (necessary) and lock (sufficient)	Winding around the origin of z	Derived
Integer value	Winding number of the locked rotor	$\pi_1(S^1) \cong \mathbb{Z}$; $W = \chi_4(q)$	Derived
Conservation	Constancy of Q , Eq. (V.2)	Abel's theorem for Wronskians [38]	Derived
Bound $ Q \leq 1$	Normalization $ z = 1$	Cauchy-Schwarz	Derived
Sign $\pm$	The two relative phases allowed by normalization	$\mathbb{Z}^\times = S^0$	Derived
Size e (that is, α)	—	—	Not derived
Coupling to A_μ	—	—	Not derived

The first five rows are topological or algebraic and use no metric. The last two rows require a scale and a coupling and are left open (Sec. VIII). A massless object has no kernels, hence no clock and no protected winding; the model therefore predicts that every massless particle is neutral and every charged particle is massive.

Table VII. Winding numbers of $\cos p\tau + i \sin q\tau$

$p \backslash q$	1	2	3	4	5	6	7
1	+1	$\times$	-1	$\times$	+1	$\times$	-1
2	0	—	0	—	0	—	0
3	+1	$\times$	—	$\times$	+1	—	-1
4	0	—	0	—	0	—	0
5	+1	$\times$	-1	$\times$	—	$\times$	-1

$\times$: the curve passes through the origin (q even) and no winding number exists. —: p and q not coprime. For odd p the entry is $\chi_4(q) = +1, -1$ for $q \equiv 1, 3 \pmod{4}$; for even p it is 0 (Appendix D).

A common misreading. “The charge must sit on one of the two kernels.”

It does not. The winding number belongs to the curve traced by the *pair* of amplitudes, not to either amplitude separately; neither kernel alone has a winding number, just as neither of two hands alone has a handshake. Assigning charge to a single term of the identity, such as the transit term [35], gives a quantity that oscillates in time and so is not conserved, and it gives detuned pairs more “charge” than locked ones. The integer is global.

What this section established. Closing the path around the origin of the amplitude plane gives an integer. For locked kernels this integer is the protected winding number $W = \pm 1$, conserved, bounded by one and carrying a sign; for detuned kernels nothing is protected. The model identifies the protected integer with charge and does not derive its size.

VI. THE QUARTIC ENERGY AS A PULLBACK OF QUADRATIC BLOCH FORMS

a. Motivation. Relativistic quantum mechanics says that energy density is quadratic in the amplitude, of the form $\psi^\dagger H \psi$, whereas the model's kernel energies are fourth powers of the amplitude. This section shows that the two are compatible: the fourth power is a quadratic energy seen through the double cover. The section keeps the results of the earlier draft of this paper and corrects its reading of temperature.

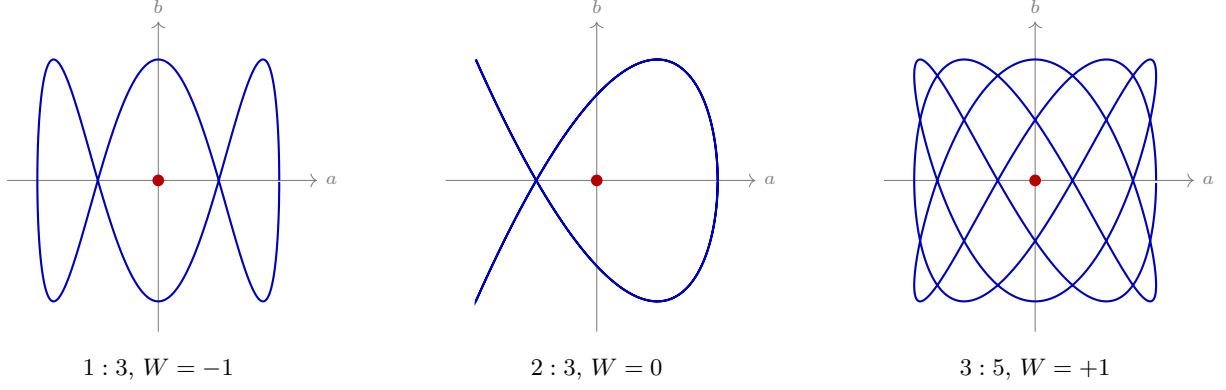


Fig. 9. Detuned pairs: windings that exist but are not protected

Each curve is $z(\tau) = \cos p\tau + i \sin q\tau$ over one period. The curves avoid the origin, so a winding number exists, with the values of Table VII. But nothing keeps them away from the origin: shifting the relative phase of the two kernels moves the curve through the origin and flips the sign of W . Compare the locked circle of Fig. 8(a), which stays at distance one from the origin for every phase. For irrational ratios the curve fills a rectangle densely and passes arbitrarily close to the origin.

A. The kernel rotor and its first-order equation

Collect the two amplitudes into a real two-component object, the *kernel rotor*,

$$\chi(t) = \begin{pmatrix} \cos \frac{\omega t}{2} \\ \sin \frac{\omega t}{2} \end{pmatrix}, \quad \chi^\dagger \chi = 1. \quad (\text{VI.1})$$

Its squared components are the occupations n_A, n_B , and it returns to itself only after $\omega t = 4\pi$, since $\chi(t + 2\pi/\omega) = -\chi(t)$. Differentiating once,

$$\dot{\chi} = \frac{\omega}{2} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \chi \implies i \partial_t \chi = H \chi, \quad H = \frac{\omega}{2} \sigma_y, \quad (\text{VI.2})$$

in units $\hbar = 1$. The generator is Hermitian with eigenvalues $\pm\omega/2$ and squares to a multiple of the identity,

$$H^2 = \left(\frac{\omega}{2}\right)^2 \mathbb{1} \implies (i\partial_t)^2 \chi = \left(\frac{\omega}{2}\right)^2 \chi. \quad (\text{VI.3})$$

[standard] The free Dirac equation [4] at rest reads $i\hbar\partial_t\psi = \beta_D mc^2\psi$ with $\beta_D^2 = \mathbb{1}$, so its square is the Klein–Gordon equation [45, 46] at rest. Equation (VI.3) is this square-root relation for a particle whose rest energy is $\hbar\omega/2$. Setting

$$\frac{\hbar\omega}{2} = mc^2 \quad (\text{VI.4})$$

makes the eigenvalues of H equal to $\pm mc^2$, the Dirac positive- and negative-energy rest energies, which Ref. [47] identified with the two kernels.

B. Unitary equivalence to the Dirac rest solutions

The eigenvectors of H are $u_\pm = (1, \pm i)^\top / \sqrt{2}$ with eigenvalues $\pm\omega/2$. Projecting and evolving gives the

exact decomposition

$$\chi(t) = \frac{1}{\sqrt{2}} \left[e^{-i\omega t/2} u_+ + e^{+i\omega t/2} u_- \right], \quad (\text{VI.5})$$

whose upper and lower components are $\cos(\omega t/2)$ and $\sin(\omega t/2)$.

a. The decomposition in steps. (i) The eigenvectors: $Hu = \lambda u$ with $H = \frac{\omega}{2}\sigma_y$ gives $\lambda = \pm\omega/2$ and $u_\pm = (1, \pm i)^\top / \sqrt{2}$. (ii) The coefficients at $t = 0$: $c_\pm = u_\pm^\dagger \chi(0) = u_\pm^\dagger (1, 0)^\top = 1/\sqrt{2}$. (iii) Each eigenvector evolves by its own phase, $e^{\mp i\omega t/2}$. (iv) Adding the two, the upper component is $\frac{1}{2}(e^{-i\omega t/2} + e^{i\omega t/2}) = \cos(\omega t/2)$ and the lower is $\frac{i}{2}(e^{-i\omega t/2} - e^{i\omega t/2}) = \sin(\omega t/2)$, which is Eq. (VI.1). No approximation enters at any step.

Theorem VI.1 (Rest-frame equivalence). *The kernel rotor $\chi(t)$ of Eq. (VI.1) is unitarily equivalent, through the fixed unitary that diagonalizes $H = \frac{\omega}{2}\sigma_y$, to the equal-weight superposition $\frac{1}{\sqrt{2}}(\psi^+ + \psi^-)$ of the Dirac positive- and negative-energy rest solutions, provided $\hbar\omega/2 = mc^2$. The kernel basis and the Dirac energy basis are one Hilbert space in two frames.*

Three consequences follow. The two branch weights are equal and constant, $|c_\pm|^2 = \frac{1}{2}$; what oscillates in the kernel frame is pure relative phase in the energy frame. The mean energy vanishes, $\langle H \rangle = 0$, so the symmetric two-kernel state is the balanced $\pm$ superposition rather than a stationary free electron. And an equal superposition of the two branches is exactly the standard condition for zitterbewegung [48, 49], so the model’s permanently oscillating interior and the standard condition for trembling motion coincide.

b. The same state in two frames. Figure 10 draws what Theorem VI.1 says. In the kernel frame the two

occupations swing between 0 and 1; in the energy frame the two branch weights stay at $\frac{1}{2}$. Nothing is lost in the change of frame: the swinging of the occupations is carried, in the energy frame, entirely by the relative phase $e^{-i\omega t}$ between the two branches. This is why the theorem can call the kernel picture and the Dirac picture “one Hilbert space in two frames.”

C. Zitterbewegung as the interference of the two branches

a. The exchange is the interference term. Substituting the decomposition (VI.5) into the occupation of kernel A , $n_A = |\chi_1|^2$, separates what each branch contributes alone from what the two contribute together,

$$n_A = \underbrace{\frac{1}{4} + \frac{1}{4}}_{\text{each branch alone}} + \underbrace{\frac{1}{2} \cos \omega t}_{\text{cross term}}, \quad (\text{VI.6})$$

and $n_B = \frac{1}{4} + \frac{1}{4} - \frac{1}{2} \cos \omega t$. The separate weights of the two branches are constant. Everything in the interior that changes in time is built from the cross term alone: the occupations, the energy flow $\dot{n}_A = -\dot{n}_B = -\frac{\omega}{2} \sin \omega t$, and the transit energy $K = \frac{1}{2} E_0 \sin^2 \omega t$. **[standard]** In Dirac theory the cross term between positive- and negative-energy solutions is the zitterbewegung [48, 49]. **[OSM reading]** The energy exchange between the two kernels is that cross term, read as a real oscillation of the electron’s interior rather than as an artefact of the representation.

b. Two readings of negative energy. **[standard]** The negative-energy solutions arise because the Dirac equation is a square root of $E^2 = p^2 c^2 + m^2 c^4$, and the square root admits both signs [4]. Quantum field theory reads the negative branch as the antiparticle. **[OSM reading]** The model reads the sign differently. An energy eigenvalue is an eigenvalue of $i\hbar \partial_t$, so its sign is the sign of a frequency: the sense in which the phase advances in time. A real oscillation such as $\cos(\omega t/2)$ contains the positive and the negative frequency with equal weight, so the real kernel amplitudes carry both branches equally within one particle, and no second particle is required. In the kernel frame the two signs register as a pair of opposite rates: while one kernel radiates, the other absorbs, $\dot{n}_A = -\dot{n}_B$ exactly. An electron with only one branch would have no cross term, hence no exchange and a frozen interior; by energy conservation the radiating side and the absorbing side must always occur together. The negative branch is therefore read not as negative energy held by anything, but as the decreasing side of a paired increase and decrease of thermal potential energy.

c. A frame statement, not a one-to-one assignment. The pairing does not assign the positive branch to kernel A and the negative branch to kernel B . The state of kernel A alone, $(1, 0)^T = (u_+ + u_-)/\sqrt{2}$, contains both branches equally. The kernel pair $\{A, B\}$ and the branch

pair $\{+, -\}$ are two bases of one two-dimensional space (Theorem VI.1); the sign pair of the energy basis appears in the kernel basis as the pair of opposite-signed rates, and the cross term between the two branches is the exchange itself.

d. What this reading must account for. Two points separate it from the standard position. First, the equal superposition has $\langle H \rangle = 0$, so the rest energy of the electron is carried by the total E_0 of Eq. (I.1), not by the mean of H (see the factor of two in Sec. X). Second, in standard theory a wave packet built from positive-energy solutions alone shows no zitterbewegung, and the Foldy–Wouthuysen transformation removes it [50]; the model’s claim is that the equal superposition is the permanent interior of the electron. The measurable consequence is the sub-luminal transport speed of Sec. IX.

A common misreading. “Without antiparticles, the negative-energy solutions of the Dirac equation have no meaning.”

In the model they keep a meaning inside one particle. The sign of an energy eigenvalue is the sign of a frequency, a real oscillation needs both, and the cross term between the two branches is exactly the energy exchange between the kernels, Eq. (VI.6). The model does not deny the antiparticle; it adds a reading of the sign that does not require one.

D. The complex unit is generated by the oscillator

a. Why this is needed. Theorem VI.1 used the imaginary unit i . If that i had to be borrowed from Dirac theory, the equivalence would be a change of basis in standard quantum mechanics rather than a statement about the model.

[standard] For any harmonic oscillator, the combination $\alpha_{\text{osc}} \propto x + ip/(m\omega)$ obeys $\dot{\alpha}_{\text{osc}} = -i\omega \alpha_{\text{osc}}$: a sinusoid is the shadow of a uniform rotation in phase space, and multiplication by i is that rotation, the symplectic complex structure [51]. **[OSM reading]** The photon sphere is driven as a harmonic oscillator by the energy gradient, with velocity and acceleration $v = \cos \omega t$, $a = -\sin \omega t$ [6]. The Pauli expectation values of the rotor are

$$\langle \sigma_z \rangle = \cos \omega t, \quad \langle \sigma_x \rangle = \sin \omega t, \quad (\text{VI.7})$$

so that $(v, a) = (\langle \sigma_z \rangle, -\langle \sigma_x \rangle)$: the oscillator’s velocity–acceleration pair traces the same circle as the Dirac zitterbewegung coherence vector, and the two share one complex structure. The identity is one of *form*; the rates of the two rotations are discussed in Sec. VII.

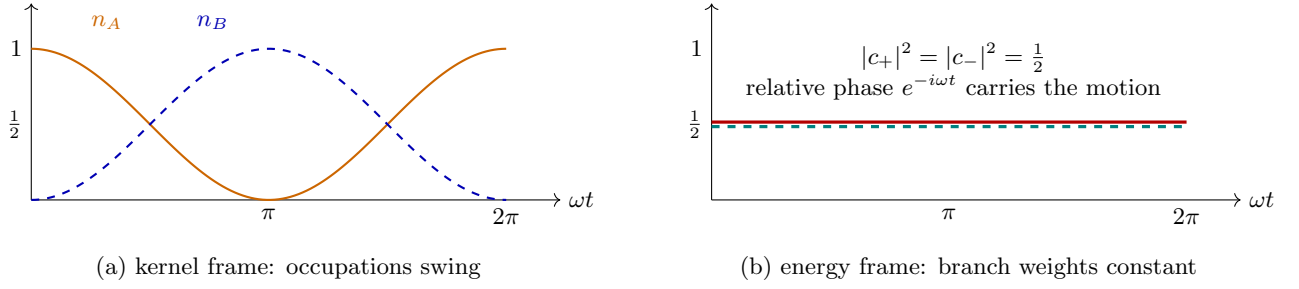


Fig. 10. One state, two frames

(a) In the kernel basis the occupations $n_A = \cos^2(\omega t/2)$ and $n_B = \sin^2(\omega t/2)$ exchange the whole energy once per half turn. (b) In the Dirac energy basis the weights of the positive- and negative-energy branches are equal and constant. The two descriptions are related by a fixed unitary transformation, Theorem VI.1.

E. The fourth power is a quadratic form on the Bloch sphere

The Bloch vector [52] $\mathbf{s} = \chi^\dagger \boldsymbol{\sigma} \chi = (\sin \omega t, 0, \cos \omega t)$ is bilinear in χ and rotates at the full angle ωt , whereas χ rotates at $\omega t/2$. The map $\chi \mapsto \mathbf{s}$ is two-to-one: it is the Hopf map [53], the double cover $SU(2) \rightarrow SO(3)$. Using $n_{A,B} = (1 \pm s_z)/2$ and $\sin^2 \omega t = s_x^2$,

$$T_A = E_0 n_A^2 = \frac{1}{4} E_0 (1 + s_z)^2, \quad (\text{VI.8})$$

$$T_B = E_0 n_B^2 = \frac{1}{4} E_0 (1 - s_z)^2, \quad (\text{VI.9})$$

$$K = \frac{1}{2} E_0 \sin^2 \omega t = \frac{1}{2} E_0 s_x^2, \quad (\text{VI.10})$$

and the conservation law becomes $\frac{1}{2}(1 + s_z^2) + \frac{1}{2}s_x^2 = \frac{1}{2}(1 + |\mathbf{s}|^2) = 1$. Every channel is quadratic in the full-angle vector $\mathbf{s}$.

a. The double cover drawn as a Riemann surface. The vector $\mathbf{s}$ is used here as bookkeeping for the kernel occupations, $s_z = n_A - n_B$, and not as a spin vector. The geometry that matters for the fourth power is the two-to-one map itself, and it is drawn without that risk as the Riemann surface of the square root, Fig. 11. [standard] The function $w = \sqrt{z}$ is two-valued; continuing it once around the branch point $z = 0$ carries it from one sheet to the other, $w \rightarrow -w$, and a second circuit returns it to the first [54]. [OSM reading] The rotor is the square root of the full-angle phase in this sense: one turn of θ carries χ to $-\chi$, on the other sheet, and two turns return it, which is the 4π periodicity of the spinor. The Dirac energy carries the same two-sheeted structure. As a function of complex momentum, $E(p) = \sqrt{p^2 c^2 + m^2 c^4}$ has branch points at $p = \pm imc$, and continuation around one of them exchanges the positive- and negative-energy sheets. The square root that linearizes the Klein–Gordon equation and the square root that relates the rotor to the occupations are thus two appearances of one kind of surface, which is why the Dirac equation and analytic continuation fit together naturally. A Riemann-surface reading of the two spin states was proposed early in the series [55].

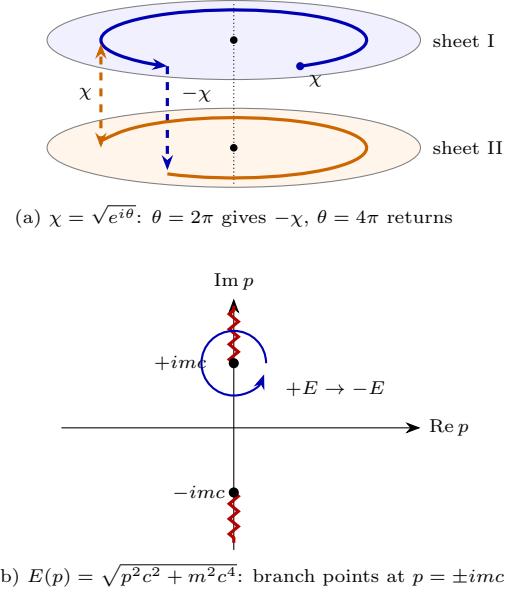


Fig. 11. The double cover as a Riemann surface

(a) The rotor as the square root of the full-angle phase. One turn of θ around the branch point (dotted axis) carries χ from sheet I to sheet II, where it reads $-\chi$; a second turn returns it. (b) The Dirac energy as a function of complex momentum. Its branch points lie at $p = \pm imc$; a loop around one of them exchanges the positive- and negative-energy sheets. Both panels show the same structure, a two-sheeted surface of the square root.

A common misreading. “The north pole of the Bloch sphere is spin up.”

Not in this paper. The north pole of $\mathbf{s}$ is the instant at which all energy is stored in kernel A; at that instant the photon sphere is at rest, $K = \frac{1}{2} E_0 s_x^2 = 0$, and nothing circulates. What the model associates with spin is not a point of this sphere but the motion along it: the circulation of the photon sphere in transit, which vanishes at the poles and is largest at the equator. Averaged over one cycle, $\langle s_x^2 \rangle = \frac{1}{2}$, the same factor $1/\sqrt{2}$ that enters the root-mean-square relation (VII.2) between β and a_e ; whether the two are one factor is left open (Table XIV).

Proposition VI.2 (Geometric origin of the quartic). *Given (i) the integrated energy gradient $T_B - T_A = -E_0 s_z$ [3], (ii) the oscillator kinetic energy $K = \frac{1}{2} E_0 s_x^2$ [6], and (iii) energy conservation, the kernel energies are uniquely $T_{A,B} = \frac{1}{4} E_0 (1 \pm s_z)^2$: quadratic forms on the Bloch sphere whose pullback through $\chi \mapsto \mathbf{s}$ is $E_0 \cos^4(\omega t/2)$ and $E_0 \sin^4(\omega t/2)$. No independent Stefan–Boltzmann assumption is used.*

Proof. Conservation and (ii) give $T_A + T_B = E_0(1 - \frac{1}{2} s_x^2) = \frac{1}{2} E_0(1 + s_z^2)$. Together with (i), $T_A = \frac{1}{2}[(T_A + T_B) - (T_B - T_A)] = \frac{1}{4} E_0(1 + s_z)^2$, and likewise for T_B . $\square$

F. Where temperature lives (correction to the earlier draft)

Reference [3] motivated the fourth powers by the Stefan–Boltzmann law [56, 57], taking the kernel temperature proportional to $\cos(\theta/2)$. The earlier draft of this paper kept that reading as a “dual” description. We withdraw it. A temperature proportional to the amplitude $\cos(\theta/2)$ would turn negative for $\theta > \pi$ and would change sign under the unobservable flip $\chi \rightarrow -\chi$. Temperature belongs to the occupation layer of Sec. ID, $\mathcal{T}_A \propto n_A$, and then the kernel energy is the *square* of temperature, $T_A = E_0 n_A^2$, the scaling of black-body radiation in one spatial dimension. The fourth power is a property of the amplitude, not of temperature. Proposition VI.2 shows where it comes from: the double cover acting on a quadratic energy.

G. Degree matching with the Dirac equation

a. Motivation. The Dirac equation in three dimensions has four solutions, and the fermionic identity is of fourth degree. The two counts are compared here factor by factor.

[standard] The number of solutions is the degree of the characteristic polynomial. For the 4×4 Dirac Hamiltonian in $3 + 1$ dimensions,

$$\det(E - H_D) = (E^2 - E_p^2)^2, \quad E_p^2 = p^2 c^2 + m^2 c^4, \quad (\text{VI.11})$$

the square of the Klein–Gordon polynomial: two distinct roots $\pm E_p$, each twice degenerate. In $1 + 1$ dimensions the Dirac matrices are 2×2 and the polynomial is $E^2 - E_p^2$ itself, with two roots. The four roots therefore factor as 2×2 . The first factor, the sign pair $\pm E_p$, comes from the square root of $E^2 = p^2 c^2 + m^2 c^4$, that is, from squaring the mass, and is present in every dimension. The second factor, the spin degeneracy, needs three spatial dimensions.

[OSM reading] The first factor is not a coincidence. The sign pair is the pair on which the model is built: the kernel pair and the branch pair are two bases of one space (Theorem VI.1), the exchange between the kernels is the

interference of the two branches (Eq. (VI.6)), and both square roots are sheets of one kind of Riemann surface (Fig. 11). The second factor remains open. It has a home in the model only if the internal angle can be identified with a spatial rotation angle, or if spin is carried by a separate degree of freedom (Sec. X). The match “fourth degree, four roots” is therefore half structural and half open.

H. Relation to the square-root construction on S^3

A companion paper [58] builds the electron’s spin two-valuedness from the square root of the Laplacian on the three-sphere $S^3 \cong SU(2)$. There the Dirac operator obeys the Lichnerowicz identity [59] $\mathcal{D}^2 = -\Delta + R/4$ with $R = 6$ on the unit sphere, its eigenvalues come in signed pairs $\pm(\ell + \frac{3}{2})$, and the obstruction to choosing one sign globally is the double cover $SU(2) \rightarrow SO(3)$. That construction is operator-theoretic and spatial. By its own account it uses no complex structure, no Bloch vector and no derivation of the quartic energy, and its Appendix C leaves open whether the square-root signature of the energy identity is “more than formal.” The present section answers that question in the affirmative: the square root appears here as the first-order evolution (VI.2) of the internal clock, the complex unit is the oscillator’s symplectic structure, and the quartic is derived (Proposition VI.2). The two papers realize the same double cover at two ends, an elliptic operator on a spatial sphere and a first-order evolution in internal time, which a Wick-rotation bridge [60] would have to join; we do not construct that bridge. The rotor is at each instant a definite point on the Bloch sphere, so the equal $\pm$ superposition of Theorem VI.1 is a deterministic rotation rather than a probabilistic mixture, consistent with the time-sequential reading of Ref. [58].

What this section established. The rotor behind the identity is a square root of Klein–Gordon at rest and is unitarily equivalent to an equal superposition of the Dirac positive- and negative-energy solutions. The complex unit is the oscillator’s own. The fourth powers are quadratic Bloch energies pulled back through the double cover; temperature lives on the occupation layer. The kernel exchange is the interference of the two Dirac branches, so the sign factor of the degree match with the Dirac equation is structural; the spin factor remains open.

VII. INTERNAL TIME: THE TRANSPORT CYCLE

a. Motivation. The angle θ advances, and the clock against which it advances has to be specified. Standard quantum mechanics uses an external time t shared by every particle in the universe. That shared time is precisely a background. A model that wants to be background independent must let each particle carry its own clock, and must say how fast it runs.

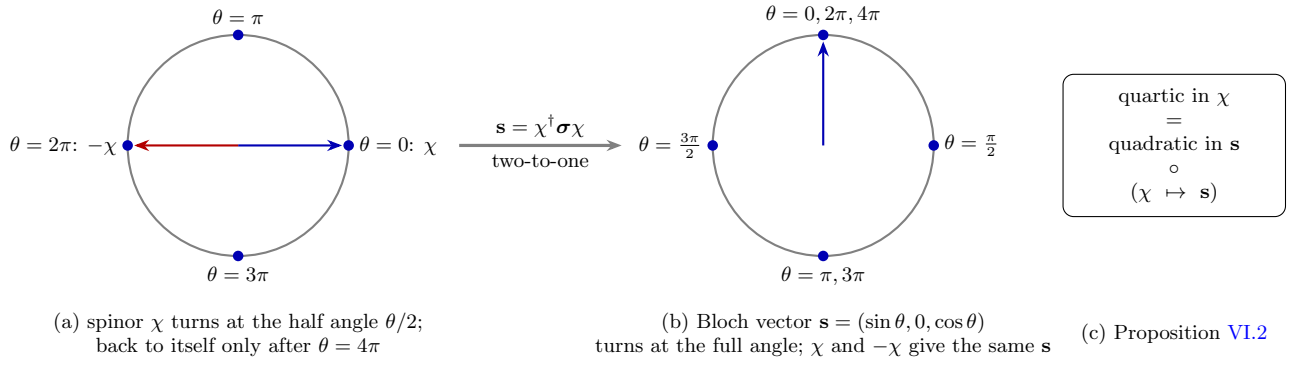


Fig. 12. The fourth power seen through the double cover

(a) The kernel rotor of Eq. (VI.1) advances at half the rate of the internal angle, so one full turn of θ brings it to $-\chi$. (b) The Bloch vector, bilinear in χ , advances at the full rate; the map $\chi \mapsto \mathbf{s}$ (the Hopf map) sends χ and $-\chi$ to the same point. (c) Each kernel energy is a quadratic form in $\mathbf{s}$; written in χ it becomes a fourth power. The fourth power is not a separate law but the quadratic energy pulled back through this two-to-one map.

A. The half-angle from kinematics

[OSM reading] Reference [6] gives the half-angle a kinematic origin. With velocity $v = \cos \omega t$ and acceleration $a = -\sin \omega t$, the Thomas precession rate [61, 62] is

$$\Omega = \frac{1}{2c^2} (\mathbf{a} \times \mathbf{v}) \propto -\frac{1}{2} \sin 2\omega t. \quad (\text{VII.1})$$

The oscillation has phase ωt , the precession has phase $2\omega t$. Writing the rotation angle as $\theta = 2\omega t$, the time phase is exactly $\theta/2$. The ratio two found here from kinematics is the same ratio that the identity requires algebraically: in $\cos^4 \phi + \sin^4 \phi + \frac{1}{2} \sin^2(k\phi) = 1$ only $k = 2$ works.

a. The computation in steps. Equation (VII.1) uses only the product of the two sinusoids. With $v = \cos \omega t$ and $a = -\sin \omega t$, and with the precession axis normal to the plane of the motion, the magnitude is $\frac{1}{2c^2} (-\sin \omega t)(\cos \omega t) = -\frac{1}{4c^2} \sin 2\omega t$, by $2 \sin x \cos x = \sin 2x$. (Reference [6] writes Ω as a scalar; the unit vector normal to the plane is supplied in Ref. [63].) The doubling of the frequency is therefore a consequence of multiplying two sinusoids of the same frequency, nothing more and nothing less.

B. The anomalous magnetic moment fixes the rate

Reference [6] reads the anomalous magnetic moment as the Lorentz contraction of the half-turn, with the root-mean-square average of a harmonic oscillation,

$$\sqrt{1 - \beta^2} = \frac{1}{1 + a_e/\sqrt{2}}, \quad (\text{VII.2})$$

$$a_e = 0.001\,159\,652\,180\,59(13),$$

which gives $\beta^2 = 0.0016379809$ and $\beta = 0.0404720$, with a_e the measured value of Ref. [64]. The form without

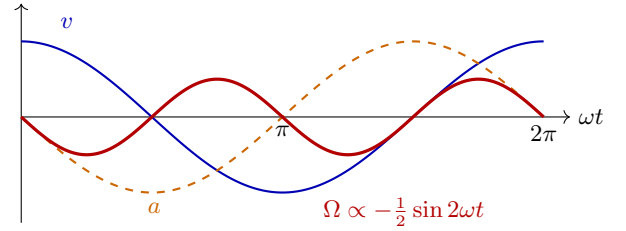


Fig. 13. Frequency doubling in the Thomas precession

With velocity $v = \cos \omega t$ and acceleration $a = -\sin \omega t$, the precession rate, proportional to their product, completes two cycles while the oscillation completes one. Read backwards: the oscillation phase is half the precession angle.

the root-mean-square factor, the bridge equation $\gamma = 1 + a_e$ [65, 66], gives the peak value 0.04812; the geometric origin of the factor $1/\sqrt{2}$ is discussed in Ref. [66]; the internal time uses the averaged rate. No parameter is adjusted: β follows from the measured a_e alone, and the resulting transport speed $\beta c \approx 0.04047 c$ is the model's principal unmeasured prediction. In standard theory a_e is computed in quantum electrodynamics, beginning with Schwinger's $\alpha/2\pi$ [67] and now to tenth order [68]; the model offers a different reading of the same measured number, not a competing calculation of it.

a. From a_e to β in four steps. (i) Divide by $\sqrt{2}$: $a_e/\sqrt{2} = 8.1999792 \times 10^{-4}$. (ii) Form the contraction factor: $1/(1 + a_e/\sqrt{2}) = 0.99918067$. (iii) Square and subtract from one: $\beta^2 = 1 - 0.99836202 = 0.0016379809$. (iv) Take the root: $\beta = 0.0404720$, so $\beta c = 1.2133 \times 10^7$ m/s. Every input is the measured a_e and the definition of c .

A common misreading. “ $0.04c$ is the speed of the electron.”

It is the speed of the internal transfer of energy between the two kernels, inside an electron that may be at rest as a whole. It is the analogue of the speed of a piston inside an engine, not the speed of the car. Dirac theory assigns the corresponding internal motion the speed of light, $\pm c$; the model’s sub-luminal value is what distinguishes it experimentally (Sec. IX).

C. Two clocks, exactly β apart

The internal oscillation now carries two rates, and the earlier draft of this paper conflated them. Table VIII separates them.

[OSM reading] We define the *internal time* of the electron as the transport cycle: one tick is one completed transfer of energy from kernel A to kernel B . In the notation of the outlook section, the tick is a map between internal states,

$$\mathcal{T}_{A \rightarrow B} : q_A^{\text{em}} \mapsto q_B^{\text{abs}}, \quad (\text{VII.3})$$

sending the state of emission at kernel A to the state of absorption at kernel B . Writing the tick as a map rather than as a phase difference $q_B - q_A$ is deliberate: a difference presupposes that the two states are read on a common scale, and supplying that scale is exactly what the transport does. The statement “time stops when radiation stops” of the series is, in this language, the statement that no tick occurs when no transfer is completed.

D. Proper time and background independence

a. Why this choice fits general relativity. The transport rate $\beta mc^2/\hbar$ is the rest energy times a pure number fixed by a_e . **[standard]** The phase of a free particle advances as $mc^2\tau/\hbar$, with τ its proper time; this Compton clock has been read out directly with atom interferometry [69]. **[OSM reading]** The internal time therefore advances as $\beta mc^2\tau/\hbar$: it is proportional to proper time, it is carried by the particle rather than by a background, in the spirit of relational accounts of time [70, 71], and it slows in a gravitational potential exactly as every atomic clock does. This last property is not optional. Local position invariance [72] has been tested with optical clocks at the level of height differences of a few tens of centimetres [73]; a clock whose rate depended on the particle species in a gravitational field would be excluded. Because β is a constant, the transport clock passes this test automatically. The path along which the transfer runs is the thermal geodesic of Refs. [22, 74], with optical metric $n^2(T)g$ [75], so the clock is also tied to geometry in the way general relativity requires.

b. A worked example: the clock one metre higher. **[standard]** Near the surface of the Earth, a clock raised by a height h runs faster by the fraction gh/c^2 , first measured in the laboratory with gamma rays [76]; for $h = 1$ m this is 1.09×10^{-16} , and optical clocks resolve it [73]. **[OSM reading]** If the internal time is proportional to proper time, the transport frequency of an electron one metre higher is larger by the same fraction: $5.0007 \times 10^{18} \text{ Hz} \times 1.09 \times 10^{-16} \approx 546 \text{ Hz}$. The point is not the number but its universality: every electron, every atom and every optical clock shifts by the same fraction, because β is a pure number. A transport clock whose shift depended on the species would contradict the observed universality of gravitational redshift. The gravitational redshift of the electron’s internal clock was first discussed in the series in Ref. [77]; the transport speed used there has since been superseded by that of Ref. [6], which does not affect a ratio of clock rates.

E. Distance from the clock

a. Motivation. So far nothing has had a length, so distance has to be built from something else. **[standard]** In the SI, since 1983 the metre has been defined from the second and the speed of light [43]: length is not an independent standard but is built from time. **[OSM reading]** The model builds the kernel separation in the same way. Rewriting $\nu_{\text{tr}} = \beta c/\lambda_C$,

$$\lambda_C = \beta c \times T_{\text{tr}}, \quad T_{\text{tr}} = \frac{1}{\nu_{\text{tr}}} = 1.9997 \times 10^{-19} \text{ s}, \quad (\text{VII.4})$$

which reproduces $\lambda_C = 2.4263 \times 10^{-12}$ m. As an equation this restates the definition in Ref. [6]; as a reading it says that the distance between the kernels is the transport speed times one tick of the internal clock. Distance is made from the clock. Building spacetime geometry from clocks and light signals alone is an old programme of general relativity [78]; here the clock is internal to the particle.

Energy then sets the distance, because $\lambda_C = hc/(mc^2)$: the electron (0.511 MeV) has $\lambda_C = 2.43 \times 10^{-12}$ m, the muon (105.7 MeV) has 1.17×10^{-14} m. The more energy, the closer the kernels. The same relation appears in the corpus as $d \times 4a = 2hc$, where $4a = 2mc^2$ is the length of the Joukowski plate on the energy axis and $d = \lambda_C$ [17]; there it was described as the place where Planck’s constant sits.

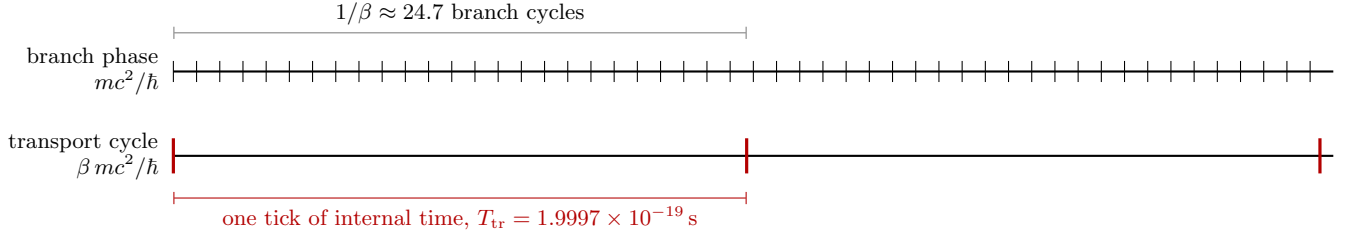
F. Frequency ledger

Table IX collects every rate that appears in the paper.

Table VIII. The two internal clocks of the electron

Clock	Angular rate	Frequency	Role
Dirac branch phase	$mc^2/\hbar$	1.2356×10^{20} Hz	Phase of each energy branch $e^{\mp imc^2 t/\hbar}$; energy layer (Sec. VI)
Transport cycle (internal $\beta mc^2/\hbar$ time)		5.0007×10^{18} Hz	One completed transfer from kernel A to kernel B; the clock of this paper

The transport frequency of Ref. [6], $\nu_{tr} = \beta c/\lambda_C$, equals $\beta mc^2/\hbar$ because $\lambda_C = \hbar/mc$. The two clocks therefore differ by exactly the factor $\beta = 0.04047$: β is the conversion rate between the energy clock and the transport clock. (Ref. [6] quotes 5.007×10^{18} Hz; recomputation with the same inputs gives 5.0007×10^{18} Hz. The difference does not affect β .)

Fig. 14. Two clocks, exactly β apart

Upper line: the phase of each Dirac energy branch, at $mc^2/\hbar = 1.2356 \times 10^{20}$ Hz. Lower line: the transport cycle from kernel A to kernel B, at 5.0007×10^{18} Hz. One transport tick lasts $1/\beta \approx 24.7$ branch cycles. The transport cycle is the internal time of this paper (Table VIII).

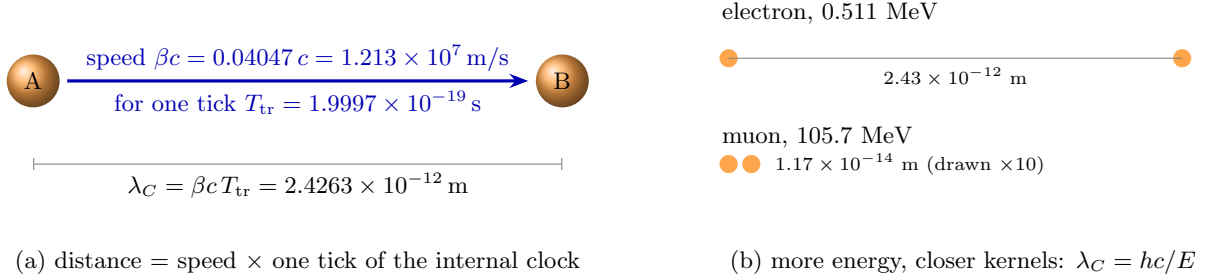


Fig. 15. Distance is made from the clock

(a) The kernel separation is the transport speed times one tick of the internal clock, Eq. (VII.4), just as the SI metre is the distance light travels in a fixed fraction of a second. (b) The separation scales inversely with the rest energy; the muon's kernels are 206.8 times closer than the electron's.

What this section established. The internal time is the transport cycle between the kernels, at the rate $\beta mc^2/\hbar$ with $\beta = 0.04047$ fixed by a_e . It runs exactly β times slower than the Dirac branch phase, is proportional to proper time, and therefore redshifts like every other clock. The kernel separation is the transport speed times one tick: distance is made from the clock.

A clock in each particle does not by itself complete background independence: one must also say how the reading of one clock is carried to another. This section is an outlook. It states what the model has, what it needs, and where each missing piece enters; it does not close the problem.

VIII. WHERE THE METRIC ENTERS: COMPARING TWO CLOCKS

a. Motivation. If every electron carries its own clock, the readings of two clocks still have to be compared.

Table IX. Frequency ledger of the internal oscillation

Rate	Carrier	Role	Layer
$\beta mc^2/\hbar$	Transport $A \rightarrow B$, Eq. (VII.3)	Internal time; one tick per completed transfer	Transport (this paper's clock)
$\omega/2 = mc^2/\hbar$	Branch phase $e^{\mp i\omega t/2}$	Single Dirac branch; the square root of the gap	Amplitude, hidden
$\omega = 2mc^2/\hbar$	Bloch vector $\mathbf{s}$; gradient $E_0 \sin \omega t$	Observable rotation; energy gap	Occupation, observable
2ω	$\Omega \propto \sin 2\omega t$; $K \propto \sin^2 \omega t$	Spin generator; transit energy	Energy (quadratic)

The last three rows are tied by the double-cover ratio 2; the first row is tied to the second by the conversion rate β . Statements that two sinusoids are “the same rotation” (Sec. VI) refer to their form; which rate a given kinematic quantity runs at is fixed by this ledger.

A. Two of Maxwell's equations need no metric

[standard] In the language of differential forms, Maxwell's equations split into

$$d\mathbf{F} = 0, \quad (\text{Faraday, no monopoles}) \quad (\text{VIII.1})$$

$$d\mathbf{G} = \mathbf{J}, \quad (\text{Gauss, Ampère}) \quad (\text{VIII.2})$$

$$\mathbf{G} = \star\mathbf{F}. \quad (\text{constitutive relation}) \quad (\text{VIII.3})$$

Here $\mathbf{F} = (\mathbf{E}, \mathbf{B})$ is the field strength and $\mathbf{G} = (\mathbf{D}, \mathbf{H})$ the excitation. Equations (VIII.1) and (VIII.2) use only the exterior derivative and need no metric; the metric, and the vacuum constants with it, enter only through the Hodge star in Eq. (VIII.3) [79]. This metric-free reading of electrodynamics goes back to Kottler, Cartan and van Dantzig [80–82]. The structural overview [5] makes the metric-free sector the starting point of the model, and Ref. [83] derives Maxwell electrodynamics as the low-energy limit of the model's line-integral ontology. We add one refinement: Eq. (VIII.2) needs only the *conservation* of charge, $d\mathbf{J} = 0$, for $\mathbf{G}$ to exist, not a mechanism for charge. The model supplies conservation (Table VI). If $\mathbf{F}$ is the curvature of a connection, Eq. (VIII.1) holds automatically.

a. The same split in vector calculus. For readers who know Maxwell's equations in vector form, Table X translates the three lines above.

B. Comparing two electrons

Keep the four kernels apart: electron 1 has kernels A, B ; electron 2 has kernels A', B' . The comparison of their clocks is a map of the same type as the tick, Eq. (VII.3), but at a different level,

$$\mathcal{T}_{(AB) \rightarrow (A'B')} : q_{AB} \mapsto q_{A'B'}^{\text{comp}}, \quad (\text{VIII.4})$$

where $q_{A'B'}^{\text{comp}}$ is the state of electron 2 assigned by comparison with electron 1, not the state of electron 2 itself. The path of the comparison is the one selected by least action, the thermal geodesic. The idea that a $U(1)$ connection is a rule for synchronizing internal clocks goes back in the series to Ref. [84], and the gauge principle itself, as a rule for comparing phases at different points, to Weyl [85]; here it is applied between electrons rather than assumed as a field. Mathematically such a map is a parallel transport [28]; if the states are phases on a circle, it is a rotation $e^{ia_{12}}$. Two outcomes are then possible. If composing comparisons around every closed chain of electrons, $\mathcal{T}_{3 \rightarrow 1} \circ \mathcal{T}_{2 \rightarrow 3} \circ \mathcal{T}_{1 \rightarrow 2}$, returns the identity, the comparisons are a pure gradient and carry no field. Products of phases around closed loops are the Wilson loops of lattice gauge theory [86], and the non-integrable phase factor is the complete description of electromagnetism in the sense of Wu and Yang [87]. If some chain does not return the identity, the comparisons have curvature: this is the rotational part of the Helmholtz decomposition (IV.1), and it is the candidate for $\mathbf{F} = (\mathbf{E}, \mathbf{B})$. Inside one electron the domain is contractible and every curl-free comparison is a gradient, so a single electron produces no holonomy; holonomy needs at least three electrons forming a loop. A continuous field A_μ is not assumed at the start: if it appears, it appears as the description of these comparisons once they fill a surface and a volume.

a. A worked example of holonomy. Suppose the comparison maps between three electrons are rotations of the clock phase by a_{12} , a_{23} and a_{31} . Going round the triangle multiplies the three rotations, so the holonomy is the phase $a_{12} + a_{23} + a_{31}$ modulo 2π . With $a_{12} = 0.3$, $a_{23} = 0.5$, $a_{31} = -0.8$ the sum is 0: the comparisons can be undone by resetting each clock's origin, and there is no field. With $a_{31} = -0.6$ instead, the sum is 0.2: no choice of clock origins removes it, because resetting the origin of electron 2 adds to a_{12} exactly what it subtracts from a_{23} .

Table X. Maxwell's equations sorted by whether they need a metric

Form language	Vector form	Metric needed	What the model supplies
$d\mathbf{F} = 0$	$\nabla \cdot \mathbf{B} = 0, \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0$	No	Automatic if $\mathbf{F}$ is the curvature of a connection (clock comparisons)
$d\mathbf{G} = \mathbf{J}$	$\nabla \cdot \mathbf{D} = \rho, \nabla \times \mathbf{H} - \partial_t \mathbf{D} = \mathbf{j}$	No	Exists if charge is conserved (winding, Table VI)
$\mathbf{G} = \star \mathbf{F}$	$\mathbf{D} = \varepsilon_0 \mathbf{E}, \mathbf{H} = \mathbf{B}/\mu_0$	Yes	Not yet: scale, light cone, α

The vector forms hide the metric inside the divergence and curl, which use lengths and angles; the form language shows that the first two lines need only the exterior derivative. The vacuum constants ε_0 and μ_0 , and with them the speed of light and the fine-structure constant, appear only in the third line.

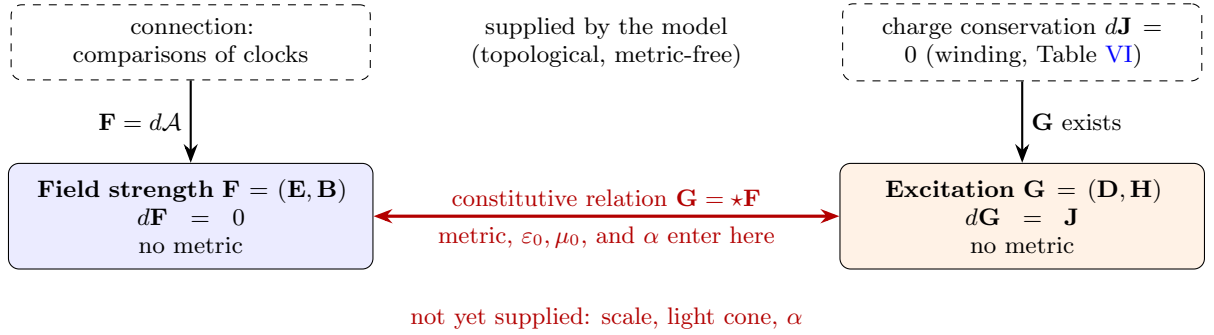


Fig. 16. Where the metric enters Maxwell's equations

Two of the equations need no metric. The left box needs only a connection, which the model would supply as the rule for comparing clocks; the right box needs only the conservation of charge, which the model supplies through the winding number. The metric and the size of the charge enter only through the constitutive relation in the middle, which is where the photon sphere's transport introduces distance (Sec. VII).

That residue is gauge-invariant; it is the discrete form of $\oint \mathcal{A} = \int \mathbf{F}$, the flux through the triangle.

b. The standard example: Aharonov–Bohm. [standard] Electrons sent around a long solenoid travel where the magnetic field is zero, yet the interference pattern shifts by the phase $e\Phi/\hbar$ [88, 89], where Φ is the flux inside the solenoid (Fig. 17). The observable is the holonomy around the loop, not a local field on the paths. The series has used this as the paradigm that line integrals, not local fields, are the primary observables [90–92], and Ref. [93] maps the route from line integrals to covariant derivatives; the comparison maps of Eq. (VIII.4) are the same idea applied between electrons.

C. From two points to a field: a research ladder

a. Why this is needed. The comparison of clocks is open, and an open problem is easier to work on when it is broken into rungs. Table XI and Fig. 19 set out the order in which the model intends to climb: begin with two electrons and a single comparison, and introduce a

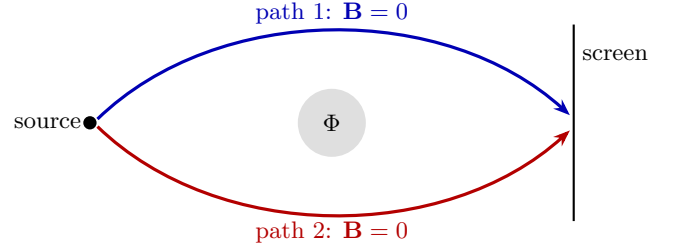


Fig. 17. Holonomy without a local field: the Aharonov–Bohm effect

Both paths avoid the magnetic field, which is confined inside the solenoid, yet the phase difference between them is $e\Phi/\hbar$. What is observed is the comparison around the closed loop.

continuous field only if the comparisons, once they fill a surface and then a volume, require one. At each rung the test is the same: whether the model's internal mechanism fixes the comparison, or something is missing.

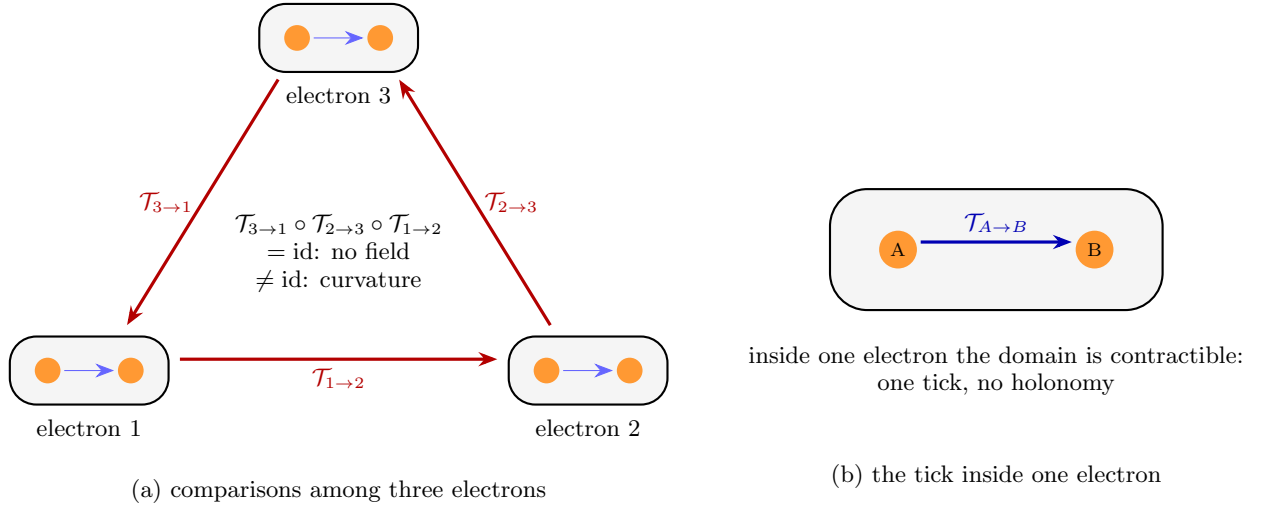


Fig. 18. One tick inside an electron, and comparisons between electrons

(a) The map $\mathcal{T}_{(AB) \rightarrow (A'B')}$ of Eq. (VIII.4), drawn between three electrons. If going round the triangle returns every clock reading to itself, the comparisons are a pure gradient and carry no field; if not, the residue is a holonomy, the candidate for $(\mathbf{E}, \mathbf{B})$. (b) The tick $\mathcal{T}_{A \rightarrow B}$ of Eq. (VII.3) is the same type of map at the lower level; the two levels are never mixed.

Table XI. The ladder from a single comparison to a field

Rung	Object	Condition tested	Mathematical name
1. Two points	One comparison $\mathcal{T}_{(AB) \rightarrow (A'B')}$ along the thermal geodesic	The comparison is fixed by the internal cycle and the path	Parallel transport
2. A chain	$\mathcal{T}_{2 \rightarrow 3} \circ \mathcal{T}_{1 \rightarrow 2}$ versus $\mathcal{T}_{1 \rightarrow 3}$	Comparison via an intermediate agrees, or not, with direct comparison	Path dependence
3. A loop	$\mathcal{T}_{3 \rightarrow 1} \circ \mathcal{T}_{2 \rightarrow 3} \circ \mathcal{T}_{1 \rightarrow 2}$	A residue survives every choice of clock origins	Holonomy
4. A surface	Many small loops tiling a surface	The residue adds up like a flux	Curvature $\mathbf{F}$
5. A volume	Surfaces closing around a region	The total flux out of a closed surface vanishes	$d\mathbf{F} = 0$

Only at rung 4 does a field-like object appear, and only if rungs 1–3 show a residue that the internal mechanism cannot remove. Rung 5 is automatic once rung 4 holds, which is why $d\mathbf{F} = 0$ needs no metric. Matching the result to the electromagnetic field additionally needs the constitutive relation and α , Table X.

D. Distance, the constitutive relation, and α enter together

Table XII collects the three layers of Eq. (IV.1) and what enters at each.

[OSM reading] While only the two kernels are placed, no metric is needed. Distance enters when the photon sphere moves, as the transport speed times one tick, Eq. (VII.4), and energy sets its scale through $\lambda_C = hc/E$. The Hodge star, which carries the metric, and the fine-structure constant, which sits inside the constitutive

relation, enter at the same place. What this supplies is a *scale*, a conformal factor. It does not yet supply the *light cone*, the directions in which signals can travel; that requires the refractive index of the thermal geodesic to be promoted from a scalar to a tensor, and it remains open.

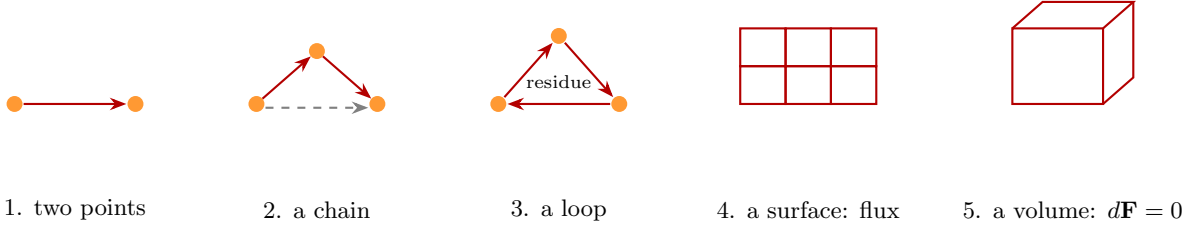


Fig. 19. Five rungs from one comparison to a field

Orange dots are electrons; red arrows are comparison maps between their internal clocks. A field-like object appears only when residues around small loops (rung 3) add up over a surface (rung 4). The model does not assume a field at the start; it asks at each rung whether one is needed.

Table XII. Three layers of the electron and what enters at each

Helmholtz part	Fluid example	0-Sphere Model	Metric needed
Gradient $\nabla\phi$	Potential flow	Kernel drive by the energy difference (Sec. IV)	No
Harmonic $\mathbf{h}$	Flow outside a point vortex	Charge as a winding number (Sec. V)	No
Rotational $\nabla \times \psi$	Flow with vorticity	Curvature of clock comparisons among three or more electrons; relation to $(\mathbf{D}, \mathbf{H})$, yes candidate for $(\mathbf{E}, \mathbf{B})$	The curvature itself, no; its

While the two kernels merely sit, everything is topological: winding numbers and closure, no lengths. Lengths enter with transport, Eq. (VII.4). At the same place enter the constitutive relation $\star$, Eq. (VIII.3), and with it the size of the charge, α . In the trunk analysis of the corpus, the origin of α and the tensor generalization of the refractive index $n(T)$ were judged to be a single open problem; the present reading locates its entrance at the moment the photon sphere crosses space.

What this section established. Two of Maxwell's equations need no metric. One needs only a connection, which the model would supply as the rule for comparing clocks; the other needs only conserved charge, which the model supplies. A single electron produces no holonomy; holonomy needs a loop of at least three electrons. Distance, the constitutive relation and α enter together when the photon sphere crosses space; the light cone is not yet supplied.

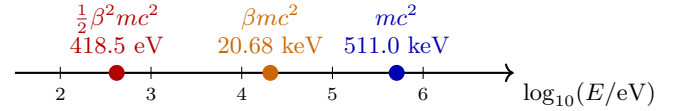


Fig. 20. Three energy scales fixed by β

Rest energy, transport quantum and transport kinetic energy of the electron, on a logarithmic axis. All three follow from a_e and mc^2 with no adjustable parameter.

IX. WHAT EXPERIMENT CAN DECIDE

a. Motivation. A paper that derives integers by topology can be internally consistent and still wrong about nature. The independent check of this paper is therefore meant to come from measurement, not from another round of argument. Table XIII lists the handles.

b. Three energy scales that follow from β . [OSM reading] Once β is fixed, three energy scales follow with no further input: the rest energy $mc^2 = 511.0$ keV; the transport quantum $\beta mc^2 = 20.68$ keV; and the kinetic energy of transport, $\frac{1}{2}mv^2 = \frac{1}{2}\beta^2 mc^2 = 418.5$ eV. Figure 20 places them on one scale. Each step down

is a factor of β or of $\beta/2$, so a measurement at any one of them constrains the others.

[OSM reading] The decisive measurement is the first row. If an internal transport speed near $0.04c$ is found, the internal clock of Sec. VII acquires an experimental footing, and with it the distance construction of Eq. (VII.4). If the speed is excluded, the transport reading of the internal time falls, while the purely topological results of Secs. II–V and the algebra of Sec. VI remain as mathematics. A measurement would not prove that a hidden variable exists; it would make the transport speed a variable of the formalism, which is how such variables have historically entered physics.

Table XIII. Experimental handles on the claims of this paper

Quantity	Value	Status	What it tests
Transport speed $v_{\text{ZB}} = \beta c$	$0.04047 c = 1.213 \times 10^7 \text{ m/s}$	Prediction, unmeasured	The sub-luminal internal motion; Dirac theory gives velocity eigenvalues $\pm c$ [48, 94]
Transport frequency ν_{tr}	$5.0007 \times 10^{18} \text{ Hz}$	Prediction, unmeasured	The internal clock (Sec. VII); Dirac's trembling frequency is $2mc^2/h = 2.47 \times 10^{20} \text{ Hz}$
Quantum $\hbar\nu_{\text{tr}} = \beta mc^2$	20.68 keV	Prediction, unmeasured	Hard X-ray window in which the transport clock would appear
Equality of electron and proton charge	$ q_e + q_p /e \lesssim 10^{-21}$ [95]	Measured, consistent	Charge as a species-blind integer: it contains no β , so it escapes the composition test of Ref. [96]
A massless charged particle	None known	Falsifier	The model predicts every massless particle is neutral (Table VI)

The first three rows are fixed by the measured a_e alone, through Eq. (VII.2), with no adjustable parameter; they are the model's principal predictions and the place where it stands or falls. The fourth row is a consistency check that the topological reading passes automatically: had the charge depended on β , which differs between species, the electron and proton charges would differ at a level long excluded. The last row is a falsifier: a single massless charged particle would contradict the statement that charge needs a clock and a clock needs mass.

X. DISTANCE FROM STANDARD THEORY

We collect the limits of the paper in one place, so that the main text could move forward without a caveat in every paragraph.

a. Rest frame only. Theorem VI.1 and Proposition VI.2 describe the zero-momentum sector. The kinetic term $c\boldsymbol{\alpha} \cdot \mathbf{p}$, Lorentz covariance and the Clifford algebra are not derived.

b. Sign sector, not spin sector. The equivalence is two-dimensional and lives in the $\pm$ energy subspace at fixed spin. The spin degeneracy of the Dirac four-spinor is a second two-valued quantity, independent of the first; where it sits in the model is open. In particular the amplitude sign $\chi \rightarrow -\chi$ cannot be the spin-up/spin-down pair, because χ and $-\chi$ are the same physical state, while spin up and spin down are orthogonal states. In the language of the Hopf fibration, the sign lives in the fibre and the spin direction in the base. The sign factor of the degree match of Eq. (VI.11) is structural (Sec. VI); the spin factor becomes structural only if the internal angle can be identified with a spatial rotation angle.

c. Charge and spin. Section V identifies the sense in which the amplitude z winds with the sign of charge. If spin is also carried by circulation, namely by the photon sphere in transit, the two quantities compete for one orientation. Either they are separate degrees of freedom, with charge the winding in the amplitude plane and spin

the angular momentum of the transported photon sphere in space, or they share one orientation, and then the model predicts a correlation between the sign of charge and the sense of spin that must be stated and tested. This paper does not decide between the two.

d. A residual factor of two. With Eq. (VI.4) the branches carry $\pm mc^2$ and the gap is $\hbar\omega = 2mc^2$. Whether E_0 in Eq. (I.1) is mc^2 or $2mc^2$ is an open bookkeeping question; Refs. [14, 15, 97] make different choices.

e. The rate of the kinematic oscillation. Section VI matches the form of the oscillator's motion with the Dirac coherence vector; Sec. VII assigns the transport its own rate, β times slower. The two statements are compatible only as a statement about form plus a ledger of rates, Table IX. Grounding the internal time in Ref. [6] adopts that paper's velocity convention, $v = \cos\omega t$.

f. Ontology. That two sinusoids are shadows of one rotation is a mathematical identity. It does not by itself establish that the internal oscillation is real; the standard position, that rest-frame trembling is a representation artifact removed by the Foldy–Wouthuysen transformation [50], remains available. The model's falsifiable separation from that position is the sub-luminal transport speed $\beta c \approx 0.04047 c$, which contradicts the $\pm c$ eigenvalues of the Dirac velocity operator; Sec. IX lists it with the other experimental handles.

g. Integers, not sizes. We repeat the statement of scope of Sec. V without qualification. The model derives the

existence, integer value, conservation, bound and sign of charge (Appendix D), and the integer structure of the two pictures. It does not derive the size of charge, the fine-structure constant α , the coupling to the electromagnetic potential, or the light-cone structure. Each is a separate open problem, named in Sec. VIII.

h. Open problems, in one table. Table XIV collects the problems this paper leaves open, with the place where each enters the argument and what would close it.

i. Spin-statistics. The two-way choice of Sec. III is reached at a different place and for a different reason than in standard theory, where it comes from the topology of exchanging identical particles. We do not claim the two are the same.

XI. CONCLUSION

We asked where the integers of an electron model come from, and followed one thread from arithmetic to physics. The integers are fixed by 0 and 1; the numbers unchanged by squaring are $\{0, 1\}$; the invertible integers are $\{\pm 1\} = S^0$. Inside the electron, the same normalization read at degree one is solved by a segment and read at degree two, in its kernel part, only by the segment's two ends: the boson is D^1 , the fermion's kernels are $S^0 = \partial D^1$, and the cross term is the seat of transport that makes the two separated points path-connected. Closing the path into a loop around the origin of the amplitude plane gives an integer winding number, which the model identifies with charge. We state the result and its limit together: the model derives the integer of charge, its existence, quantization, conservation, bound and sign, and it does not derive the size of the charge; the fine-structure constant remains an input, exactly as in quantized vortices and flux quanta, where the integer and its quantum come from different places.

The earlier content of this paper is retained: the rotor behind the identity is a square root of Klein–Gordon at rest and is unitarily equivalent to the Dirac rest solutions; the complex unit is the oscillator's symplectic structure; and the fourth powers are quadratic Bloch forms pulled back through the Hopf double cover, with temperature now placed in the occupation layer. The internal time is the transport cycle, running at $\beta mc^2/\hbar$ with $\beta = 0.04047$ fixed by the anomalous magnetic moment, exactly β times slower than the Dirac branch phase and proportional to proper time; the distance between the kernels is the transport speed times one tick. No metric is needed while the kernels merely sit; distance, the constitutive relation and the fine-structure constant enter together when the photon sphere crosses space. The model stands or falls experimentally on the transport speed $\beta c \approx 0.04c$, fixed by a_e with no free parameter. Figure 21 summarises the flow.

ACKNOWLEDGMENTS

The author used a large language model (LLM) as a writing aid during the preparation of this manuscript, in a supporting role only: to help organize the presentation of the text and to check grammar and wording. The physical ideas, the arguments, the derivations, and the structure of the paper and of its text are the author's own, and the author takes full responsibility for the entire content.

Appendix A: Numerical Verification

a. What the script does, and why. Every number and every identity in this paper can be checked by a short Python script, `verify65.py`, distributed with the paper. The script contains seven independent checks. Each check computes a quantity in two ways, or tests an inequality on a fine grid, and stops with an error if the two disagree; if all seven pass it prints the values of β and the transport frequency and the line `all checks passed`. The script uses only `numpy` [98] and the Python standard library and runs in under a second. Figure 22 shows which claim of the paper each check supports, and the listings below give each check with a one-sentence statement of its purpose.

b. Setup. The Pauli matrices, the rotor $\chi(t)$ of Eq. (VI.1), and units $\omega = E_0 = 1$.

```
1 import numpy as np, math
2 sx=np.array([[0,1],[1,0]]); sy=np.array([[0,-1j],
3     [1j,0]]); sz=np.array([[1,0],[0,-1]])
4 w=1.0; E0=1.0
5 chi=lambda t: np.array([np.cos(w*t/2), np.sin(w*
6     t/2)], dtype=complex)
```

c. Check 1: two solution sets. On a grid of 10^5 points of the segment, the kernel terms $n_A^p + n_B^p$ equal one at the ends and stay below one inside, for $p = 1.5, 2, 3$; and the cubic $\binom{p}{3} \propto p^3 - 3p^2 + 2p$ has roots $\{0, 1, 2\}$.

```
1 x=np.linspace(0,1,100001)
2 for p in [1.5,2,3]:
3     f=x**p+(1-x)**p
4     assert np.all(f[1:-1]<1) and f[0]==1 and f
5     [-1]==1
6 assert sorted(np.round(np.roots([1,-3,2,0]).real
7     ,12))==[0,1,2]
```

d. Check 2: the first-order equation. The numerical derivative of χ satisfies $i\dot{\chi} = H\chi$ with $H = \frac{\omega}{2}\sigma_y$, whose eigenvalues are $\pm\omega/2$.

```
1 dt=1e-6
2 for t in np.linspace(0.1,5,7):
3     d=(chi(t+dt)-chi(t-dt))/(2*dt)
4     assert np.max(np.abs(1j*d-(w/2)*sy@chi(t)))
5     <1e-6
6 assert np.allclose(np.linalg.eigvalsh((w/2)*sy)
7     ,[-0.5,0.5])
```

Table XIV. Open problems and where each enters

Problem	Where it enters	What would close it
Size of charge, α	Constitutive relation, Eq. (VIII.3)	A derivation of the Hodge star, with its scale, from the transport of the photon sphere
Coupling to A_μ	Comparison of clocks, Sec. VIII	Rungs 1–4 of Table XI carried out from the internal mechanism
Light cone	Refractive index of the thermal geodesic	Promotion of the scalar index $n(T)$ to a tensor that fixes the causal directions
Spin sector	Second two-valued quantity of the Dirac spinor	An identification of the internal angle with a spatial rotation angle, or another home for spin up/down
Charge and spin	Sense of rotation of z , Sec. V	A decision whether charge and spin are separate degrees of freedom or share one orientation
Origin of $1/\sqrt{2}$	Eq. (VII.2); cycle average of the transit energy	A demonstration that $\langle s_x^2 \rangle = \frac{1}{2}$ of the circulating photon sphere is the root-mean-square factor of Ref. [66]
Factor of two in E_0	Eq. (VI.4)	A decision whether E_0 is mc^2 or $2mc^2$, consistent across the series
Velocity convention	Sec. VII	Confirmation of the convention $v = \cos \omega t$ of Ref. [6]
Reality of the internal motion	Sec. IX	A measurement of the transport speed near $0.04 c$, or its exclusion

The first three rows are one problem seen from three sides: the scale, the coupling and the causal structure all enter where the photon sphere crosses space. The last row is experimental and is the decisive one.

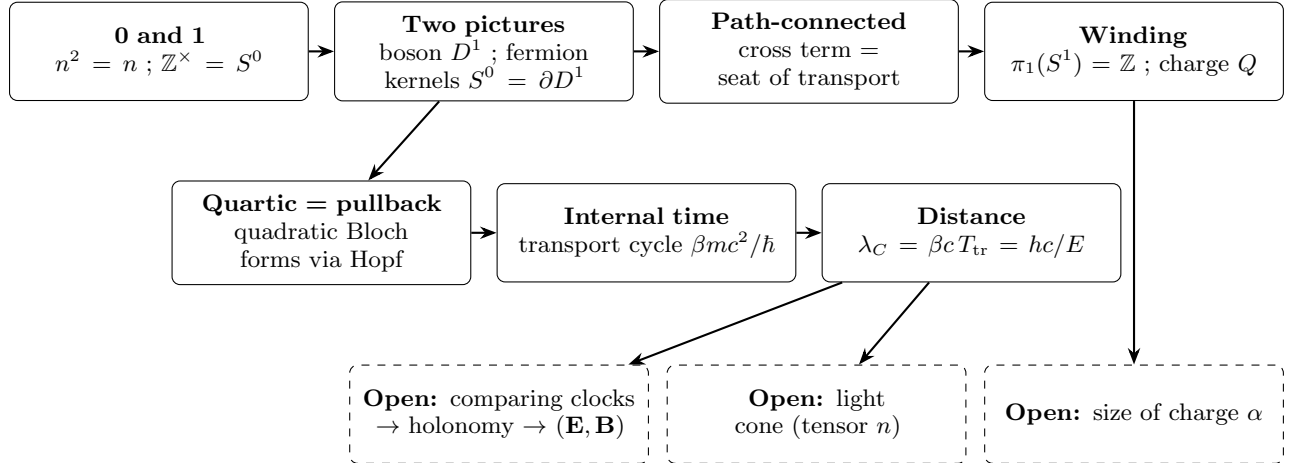


Fig. 21. The flow of the argument

Top row: the integer thread, from the arithmetic of 0 and 1 (Sec. II) through the two solution sets (Sec. III) and path-connectedness (Sec. IV) to the winding number identified with charge (Sec. V). Middle row: the energy and time thread, from the quartic as a Hopf pullback (Sec. VI) to the internal time and to distance built from it (Sec. VII). Bottom row, dashed: what is not derived here; the size of the charge hangs below the winding thread, while clock comparison and the light cone hang below distance, where the metric enters (Sec. VIII).

e. Check 3: the two Dirac branches. Projected on the eigenvectors of H , the rotor has weight $\frac{1}{2}$ on each branch at every instant, and the mean energy vanishes. The

occupation of kernel A splits into $\frac{1}{4} + \frac{1}{4}$ from the branches alone and the cross term $\frac{1}{2} \cos \omega t$, Eq. (VI.6).

```
1 vals, V=np.linalg.eigh((w/2)*sy)
```

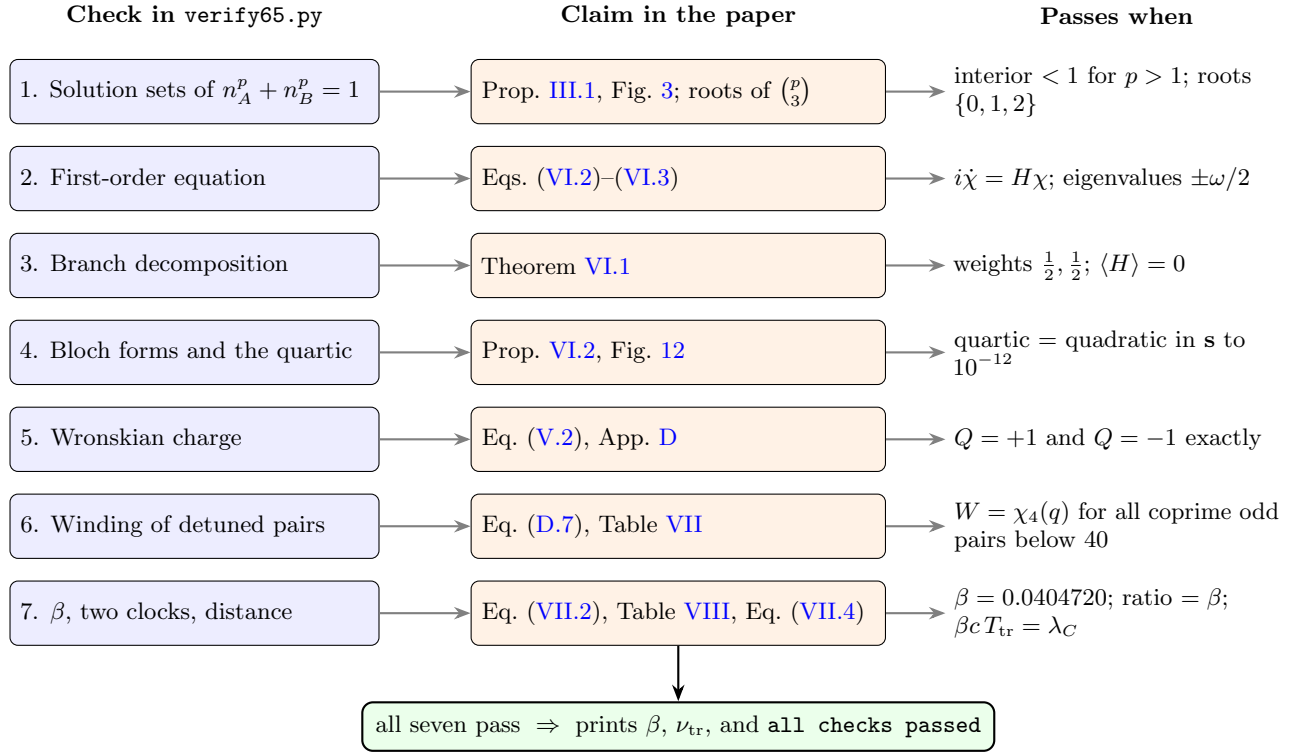


Fig. 22. Map of the numerical checks

Each row is one block of the script. Left: what the block computes. Middle: the statement of the paper it supports. Right: the condition under which the block passes. The checks are independent of one another; any single failure stops the script with an assertion error that names the line.

```

2 for t in np.linspace(0, 2*np.pi, 9):
3   c = V.conj().T@chi(t)
4   assert np.allclose(np.abs(c)**2, [0.5, 0.5])
5   assert abs((chi(t).conj()@((w/2)*sy)@chi(t)).real) < 1e-12
6   a = c[0]*V[:, 0]; b = c[1]*V[:, 1]
7   assert abs(abs(a[0])**2 + abs(b[0])**2 - 0.5) < 1e-12
8   assert abs(2*(a[0]*b[0].conjugate()).real - 0.5*np.cos(w*t)) < 1e-12

```

f. Check 4: the quartic as a pullback. At 50 instants, the quartic kernel energies equal the quadratic Bloch forms, and solving the sum (conservation and kinetic energy) and the difference (energy gradient) recovers $\cos^4(\omega t/2)$.

```

1 for t in np.linspace(0, 2*np.pi, 50):
2   c = chi(t); Sx = (c.conj()@sx@c).real; Sz = (c.conj()@sz@c).real
3   assert abs(E0*np.cos(w*t/2)**4 - E0*((1+Sz)/2)**2) < 1e-12
4   assert abs(0.5*E0*np.sin(w*t)**2 - 0.5*E0*Sx**2) < 1e-12
5   KE = 0.5*E0*Sx**2; s = E0 - KE; d = -E0*Sz
6   assert abs((s-d)/2 - E0*np.cos(w*t/2)**4) < 1e-12

```

g. Check 5: the Wronskian charge. For the rotor, $Q = (2/\omega)(ab - ba)$ equals $+1$ at every instant, and -1 when

$b \rightarrow -b$.

```

1 t = np.linspace(0, 10, 1001); a = np.cos(w*t/2); b = np.sin(w*t/2)
2 ad = -(w/2)*np.sin(w*t/2); bd = (w/2)*np.cos(w*t/2)
3 assert np.allclose((2/w)*(a*bd - b*ad), 1)
4 assert np.allclose((2/w)*(a*(-bd) - (-b)*ad), -1)

```

h. Check 6: winding of detuned pairs. The signed-crossing sum (D.6) equals the mod-4 character $\chi_4(q)$ for every coprime pair of odd p, q below 40.

```

1 def chi4(n): return 0 if n%2==0 else (1 if n%4==1 else -1)
2 def W(p, q): return 0.5*sum((-1)**k*math.copysign(1,
3   math.sin(math.pi*q*(2*k+1)/(2*p))) for k in range(2*p))
4 for p in range(1, 40, 2):
5   for q in range(1, 40, 2):
6     if math.gcd(p, q) == 1: assert W(p, q) == chi4(q)

```

i. Check 7: β , the two clocks, and distance. From the measured a_e , compute β by Eq. (VII.2); confirm that the transport frequency is β times the branch frequency and that the transport speed times one tick is the Compton wavelength.

```

1 ae = 0.00115965218059

```

```

2 beta=math.sqrt(1-1/(1+ae/math.sqrt(2))**2)
3 h=6.62607015e-34; c=299792458; me=9.1093837015e
  -31
4 lamC=h/(me*c); nu_branch=me*c**2/h; nu_tr=beta*c
  /lamC
5 assert abs(beta-0.0404720)<1e-7
6 assert abs(nu_tr/nu_branch-beta)<1e-15
7 assert abs(beta*c/nu_tr-lamC)<1e-25
8 print(f"beta={beta:.7f} nu_tr={nu_tr:.5e} Hz")
9 print("all checks passed")

```

j. Expected *output.* beta=0.0404720
 nu_tr=5.00068e+18 Hz followed by all checks
 passed.

Appendix B: The Bloch Map and the Derivation of the Quartic

a. Half-angle to full-angle. For the rotor (VI.1), the occupations are $n_{A,B} = \frac{1}{2}(1 \pm \cos \omega t)$ by the half-angle formulas, so $s_z = n_A - n_B = \cos \omega t$ and, from $2 \cos \phi \sin \phi = \sin 2\phi$, $s_x = \sin \omega t$; $s_y = 0$ because χ is real. The rotor advances by $\omega t/2$, the Bloch vector by ωt : this is the two-to-one map.

b. Uniqueness. The two equations of the proof of Proposition VI.2, the sum fixed by conservation and the oscillator energy and the difference fixed by the energy gradient, determine T_A and T_B separately. The coefficient structure $\frac{1}{4}(1 \pm s_z)^2$ shows that the fourth power in χ is the square of an occupation that is linear in s_z .

c. The boson in the same language. Squaring the bosonic identity (III.1) at the half angle reproduces the fermionic identity, with cross term $2 \sin^2(\theta/2) \cos^2(\theta/2) = \frac{1}{2} \sin^2 \theta$. The fourth power and the transit seat are what the second squaring adds; the sign flip under $\theta \rightarrow \theta + 2\pi$ is already present at the first.

Appendix C: Words That Split in Two

Several words borrowed from established fields carry two meanings in this paper. When a reader's objection seems to miss the point, the first thing to check is whether the same word is being read in the other sense.

a. Occupation number. Here: the share $n_A = \cos^2(\theta/2)$ of the energy held by kernel A, a continuous quantity. In standard theory: the eigenvalue 0 or 1 of the fermion number operator. Both satisfy the idempotent equation (II.1) at the ends, but they are different objects.

b. Proper time. Here: the internal clock carried by the particle, the transport cycle. In general relativity: the length of a worldline given by the metric. Section VII argues that the first runs proportionally to the second, which is a claim, not a definition.

c. Radiation. Here: the transfer of energy between the two kernels. In electromagnetism: emission of electromagnetic waves to infinity. "Time stops when radiation stops" refers to the first.

d. Double cover. Here: the sign flip of the real amplitude after one turn. In standard theory also: the covering $SU(2) \rightarrow SO(3)$ of spatial rotations. Whether the two coincide is the open question of Sec. X.

e. Hidden variable. Here: a degree of freedom absent from the standard formalism, carried by the particle, as density is carried by a particle in smoothed-particle hydrodynamics [99]. In the Bell literature [100]: a local variable meant to explain two-particle correlations. This paper uses only the first sense.

Appendix D: Derivation: Charge as a Protected Winding Number

This appendix derives the five rows of Table VI marked *derived*. Everything here is elementary: trigonometric identities, the winding number of a closed curve, and Abel's theorem for Wronskians. No metric and no coupling constant is used, which is why the size of the charge cannot come out of it.

1. The amplitude plane and the winding number

Let $a(t)$ and $b(t)$ be the amplitudes of kernels A and B, and write $z = a + ib$. The occupations are $n_A = a^2$, $n_B = b^2$, so $|z|^2 = n_A + n_B$. [standard] If a closed curve $z(t)$, $0 \leq t \leq T$, avoids the origin, its winding number [27]

$$W = \frac{1}{2\pi} \oint d \arg z = \frac{1}{2\pi} \int_0^T \frac{a \dot{b} - b \dot{a}}{a^2 + b^2} dt \quad (D.1)$$

is an integer, and it cannot change under any continuous deformation of the curve that keeps away from the origin. It can change only when the curve passes through the origin, that is, when both kernels are empty at the same instant. This is why two kernels are *necessary*: a single amplitude passes through zero twice per cycle [3, 29].

2. Normalization forces lock and selects two orientations

Consider two kernels at the same frequency with general amplitudes and relative phase, $a = A \cos t$, $b = B \sin(t + \delta)$. A direct expansion gives

$$a^2 + b^2 = \frac{1}{2}(A^2 + B^2) + \frac{1}{2}(A^2 - B^2 \cos 2\delta) \cos 2t + \frac{1}{2}B^2 \sin 2\delta \sin 2t. \quad (D.2)$$

The normalization $n_A + n_B = 1$ at every instant requires both oscillating coefficients to vanish: $\sin 2\delta = 0$ and $A^2 =$

$B^2 \cos 2\delta$. With $B \neq 0$ the second condition excludes $\cos 2\delta = -1$, so

$$A = B = 1, \quad \delta \in \{0, \pi\}. \quad (\text{D.3})$$

For $\delta = 0$, $z = e^{it}$ and $W = +1$; for $\delta = \pi$, $z = e^{-it}$ and $W = -1$. The identity therefore has exactly two locked solutions, distinguished by orientation; the model reads them as e^- and e^+ . Any intermediate δ gives an ellipse with $|z| < 1$ somewhere, and $\delta = \pi/2$ collapses the curve to the segment $a = b$ through the origin (Fig. 8c).

For $A = B = 1$ the oscillating part of Eq. (D.2) is $\frac{1}{2}(1 - \cos 2\delta) \cos 2t + \frac{1}{2} \sin 2\delta \sin 2t$, whose amplitude is $\frac{1}{2} \sqrt{(1 - \cos 2\delta)^2 + \sin^2 2\delta} = \frac{1}{2} \sqrt{2 - 2 \cos 2\delta} = |\sin \delta|$. Hence $\min_t |z|^2 = 1 - |\sin \delta|$, which is Eq. (V.1): the gap closes continuously as δ moves from 0 to $\pi/2$ (Fig. 7). Two kernels running at different frequencies cannot satisfy the normalization at all, since the cross term then oscillates at the beat frequency. Normalization is therefore equivalent to lock.

3. The Wronskian: conservation, quantization, bound

Define, for a pair at internal angular frequency ω ,

$$Q = \frac{2}{\omega} (a \dot{b} - b \dot{a}). \quad (\text{D.4})$$

a. Conservation. If a and b oscillate at angular frequencies ω_A and ω_B , so that $\ddot{a} = -\omega_A^2 a$ and $\ddot{b} = -\omega_B^2 b$, then

$$\frac{d}{dt} (a \dot{b} - b \dot{a}) = \ddot{a} b - a \ddot{b} = (\omega_A^2 - \omega_B^2) ab. \quad (\text{D.5})$$

For locked kernels the right-hand side vanishes identically: the Wronskian of two solutions of the same linear second-order equation is constant (Abel) [38]. For detuned kernels it does not.

b. Quantization. For the locked rotor $a = \cos(\omega t/2)$, $b = \pm \sin(\omega t/2)$, Eq. (D.4) gives $Q = \pm 1$ exactly.

c. Bound. By the Cauchy–Schwarz inequality $|a \dot{b} - b \dot{a}| \leq |z| |\dot{z}|$. On the normalized circle $|z| = 1$ and $|\dot{z}| = \omega/2$, so $|Q| \leq 1$, with equality only when $\dot{z} \perp z$, that is, for circular motion. There is no doubly charged locked pair.

d. Winding. Comparing Eqs. (D.1) and (D.4), the Wronskian is the density of the winding number: on the unit circle, W is the time integral of $(\omega/2)Q$ divided by 2π . Charge as an integer and charge as a conserved current are the same quantity, integrated or not.

4. Detuned kernels: the mod-4 character

For coprime frequency ratios $p : q$, take the curve $z(\tau) = \cos p\tau + i \sin q\tau$, $0 \leq \tau \leq 2\pi$. If q is even, the curve passes through the origin and W is undefined. Otherwise the curve crosses the imaginary axis at $\tau_k = \pi(2k+1)/(2p)$, $k = 0, \dots, 2p-1$, in the direction $(-1)^k$ and on the side sign $\sin q\tau_k$; counting signed crossings of the positive imaginary axis gives

$$W = \frac{1}{2} \sum_{k=0}^{2p-1} (-1)^k \text{sign} \sin \frac{\pi q(2k+1)}{2p}. \quad (\text{D.6})$$

Write $n = 2k+1$ and substitute $n \equiv qm \pmod{4p}$; for odd q coprime to p this permutes the odd residues. With the mod-4 Dirichlet character [34] $\chi_4(m) = 0, +1, -1$ for m even, $m \equiv 1, m \equiv 3 \pmod{4}$, one has $(-1)^{(m-1)/2} = \chi_4(m)$ and $\chi_4(q^{-1}n) = \chi_4(q)\chi_4(n)$, while $\text{sign} \sin(\pi n/2p)$ is $+1$ for $0 < n < 2p$ and -1 for $2p < n < 4p$. Hence

$$W = \frac{1}{2} \chi_4(q) \left[\sum_{\substack{n \text{ odd} \\ n < 2p}} \chi_4(n) - \sum_{\substack{n \text{ odd} \\ 2p < n < 4p}} \chi_4(n) \right]. \quad (\text{D.7})$$

For odd p the first sum has p alternating terms beginning with $+1$ and equals 1; the second begins at $2p+1 \equiv 3 \pmod{4}$ and equals -1 ; so $W = \chi_4(q)$. For even p both sums have an even number of alternating terms and vanish, so $W = 0$. The winding number of any coprime detuned pair is therefore 0 or ± 1 , never larger, and it is not protected: changing the relative phase moves the curve through the origin and flips its sign. Direct numerical counting agrees with Eq. (D.7) for every coprime odd pair up to 301. For irrational ratios the curve comes arbitrarily close to the origin and no winding number is defined. Detuned kernels carry no conserved integer: the model reads them as neutral.

5. What this appendix does not show

Nothing above fixes the size of one unit of charge or shows that the winding number couples to an electromagnetic potential. Those require a scale and a coupling, the two rows of Table VI marked *not derived*.

Appendix E: Mathematical Terms in Plain Words

This appendix gives a one-sentence meaning for every mathematical term the paper uses, for readers coming from outside mathematical physics.

Table XV. Mathematical terms used in the paper

Term	Plain meaning	Where it is used
Idempotent	A number unchanged by squaring; among the reals, only 0 and 1	Secs. II, III
Zero-sphere S^0	The set of two points at unit distance from the origin of a line	Throughout
Path-connected	Any two points can be joined by a continuous path inside the set	Sec. IV
Simply connected	Every closed loop can be shrunk to a point inside the set; a circle is not	Sec. V
Winding number	The number of times a closed curve goes round a point, counted with sign	Sec. V, App. D
Wronskian	$a\dot{b} - b\dot{a}$; constant for two solutions of the same linear oscillator equation	Eq. (V.2)
Dirichlet character χ_4	The function 0, +1, -1 on even numbers, numbers $\equiv 1 \pmod{4}$ and $\equiv 3 \pmod{4}$	Eq. (D.7)
Helmholtz (Hodge) decomposition	Splitting a flow into gradient, rotational and harmonic parts	Eq. (IV.1)
Double cover; Hopf map	A two-to-one map; here the map from the spinor χ to the Bloch vector $\mathbf{s}$	Sec. VI
Riemann surface; branch point	The surface on which a many-valued function such as $\sqrt{z}$ becomes single-valued; going once around a branch point moves to the other sheet	Fig. 11
Interference term	(cross) The part of $ a + b ^2$ that needs both a and b , namely $2\text{Re}(a\bar{b})$	Eq. (VI.6)
Symplectic structure	complex The rotation of position–momentum phase space by a quarter turn; the “ i ” of an oscillator	Sec. VI
Connection; transport	parallel A rule for carrying a quantity from one place to another so that it can be compared	Sec. VIII
Holonomy	What remains after carrying a quantity round a closed loop and comparing it with itself	Sec. VIII
Hodge star $\star$	The operation that needs a metric and turns field strength into excitation	Eq. (VIII.3)
Lock; detuning	Two kernels at the same frequency; at different frequencies	Sec. V

The definitions are informal by design. Each term is used in the paper only in the sense given here.

- [1] G. E. Uhlenbeck and S. Goudsmit, *Spinning Electrons and the Structure of Spectra*, *Nature* **117**, 264 (1926).
- [2] J. J. Sakurai and J. Napolitano, *Modern Quantum Mechanics*, 2nd ed. (Addison-Wesley, San Francisco, 2011).
- [3] S. Hanamura, *A Model of an Electron Including Two Perfect Black Bodies*, Zenodo (2018), [10.5281/zenodo.16759284](https://zenodo.org/record/16759284).
- [4] P. A. M. Dirac, *The Quantum Theory of the Electron*, *Proc. R. Soc. Lond. A* **117**, 610 (1928).
- [5] S. Hanamura, *The 0-Sphere Model: A Structural Overview*, Zenodo (2026), in preparation.
- [6] S. Hanamura, *Redefining Electron Spin and Anomalous Magnetic Moment Through Harmonic Oscillation and Lorentz Contraction*, Zenodo (2024), [10.5281/zenodo.17764997](https://zenodo.org/record/17764997).
- [7] S. Lang, *Algebra*, revised 3rd ed. (Springer, New York, 2002).
- [8] P. A. M. Dirac, *Quantised Singularities in the Electromagnetic Field*, *Proc. R. Soc. Lond. A* **133**, 60 (1931).
- [9] F. London, *Superfluids*, Vol. I (Wiley, New York, 1950).
- [10] B. S. Deaver and W. M. Fairbank, *Experimental Evidence for Quantized Flux in Superconducting Cylinders*, *Phys. Rev. Lett.* **7**, 43 (1961).
- [11] R. Doll and M. Näbauer, *Experimental Proof of Magnetic Flux Quantization in a Superconducting Ring*, *Phys. Rev. Lett.* **7**, 51 (1961).
- [12] K. v. Klitzing, G. Dorda, and M. Pepper, *New Method for High-Accuracy Determination of the Fine-Structure Constant Based on Quantized Hall Resistance*, *Phys. Rev. Lett.* **45**, 494 (1980).
- [13] D. J. Thouless, M. Kohmoto, M. P. Nightingale, and M. den Nijs, *Quantized Hall Conductance in a Two-Dimensional Periodic Potential*, *Phys. Rev. Lett.* **49**, 405 (1982).
- [14] S. Hanamura, *Electron Interference from Internal Geometry: Two-Kernel SU(2) Structure, Quartic Energy Flow, and an Intrinsic Visibility Limit*, Zenodo (2026), [10.5281/zenodo.18718174](https://zenodo.org/record/18718174).
- [15] S. Hanamura, *Geometric Origin of the One-Half Factor: Thermal Potential Energy Floor and Zero-Point Energy in the 0-Sphere Model*, Zenodo, [10.5281/zenodo.19010945](https://zenodo.org/record/19010945).
- [16] J. Milnor, *Morse Theory* (Princeton University Press, Princeton, 1963).
- [17] S. Hanamura, *0-Sphere Model Working Notes: Trunks T1–T7 (Identity Algebra, Quantum Numbers, β Hierarchy, Internal Dynamics, Interaction Picture, Detuned Kernels, Two Particles)*, GitHub repository HanaTensor/0sm-skills, directory `trunks` (2026), github.com/HanaTensor/0sm-skills.
- [18] W. Pauli, *The Connection Between Spin and Statistics*, *Phys. Rev.* **58**, 716 (1940).
- [19] D. Finkelstein and J. Rubinstein, *Connection between Spin, Statistics, and Kinks*, *J. Math. Phys.* **9**, 1762 (1968).
- [20] J. M. Leinaas and J. Myrheim, *On the Theory of Identical Particles*, *Nuovo Cimento B* **37**, 1 (1977).
- [21] S. Hanamura, *Geometrical Confinement of Energy in the 0-Sphere Model: A Topological Mechanism Underlying Rest Mass and Zitterbewegung*, Zenodo (2026), [10.5281/zenodo.18356895](https://zenodo.org/record/18356895).
- [22] S. Hanamura, *Helical Trajectory, Fixed-Endpoint Line Integrals, and the Emergence of the Spacetime Metric in the 0-Sphere Model*, Zenodo (2026), [10.5281/zenodo.20388056](https://zenodo.org/record/20388056).
- [23] S. B. Myers, *Riemannian Manifolds with Positive Mean Curvature*, *Duke Math. J.* **8**, 401 (1941).
- [24] H. Helmholtz, *Über Integrale der hydrodynamischen Gleichungen, welche den Wirbelbewegungen entsprechen*, *J. Reine Angew. Math.* **55**, 25 (1858).
- [25] H. Bhatia, G. Norgard, V. Pascucci, and P.-T. Bremer, *The Helmholtz–Hodge Decomposition—A Survey*, *IEEE Trans. Vis. Comput. Graph.* **19**, 1386 (2013).
- [26] W. V. D. Hodge, *The Theory and Applications of Harmonic Integrals* (Cambridge University Press, Cambridge, 1941).
- [27] A. Hatcher, *Algebraic Topology* (Cambridge University Press, Cambridge, 2002).
- [28] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed. (Institute of Physics, Bristol, 2003).
- [29] S. Hanamura, *Emergent Conservation Laws from Internal Geometry: A Noetherian Reinterpretation in the 0-Sphere Model*, Zenodo (2025), [10.5281/zenodo.17765244](https://zenodo.org/record/17765244).
- [30] S. Hanamura, *Rotation from Scalar Oscillation: Emergence of Photon-Sphere Angular Momentum from Phase-Staggered Kernel Dynamics in the 0-Sphere Model*, Zenodo (2026), [10.5281/zenodo.19482145](https://zenodo.org/record/19482145).
- [31] G. K. Batchelor, *An Introduction to Fluid Dynamics* (Cambridge University Press, Cambridge, 1967).
- [32] P. G. Saffman, *Vortex Dynamics* (Cambridge University Press, Cambridge, 1992).
- [33] W. Thomson (Lord Kelvin), *On Vortex Motion*, *Trans. R. Soc. Edinburgh* **25**, 217 (1869).
- [34] T. M. Apostol, *Introduction to Analytic Number Theory* (Springer, New York, 1976).
- [35] S. Hanamura, *On the Non-Perturbative Nature of the 0-Sphere Model and Magnetic Monopole Absence: A Supplement on Structural Analysis and Physical Interpretation*, Zenodo (2026), [10.5281/zenodo.18857609](https://zenodo.org/record/18857609).
- [36] S. Hanamura, *On Internal Energy Stabilization, Charge Localization, and Superradiant Rephasing: Supplementary Note on Geometric Stability Mechanisms in the 0-Sphere Model*, Zenodo (2026), [10.5281/zenodo.18905344](https://zenodo.org/record/18905344).
- [37] S. Hanamura, *Geometric Origin of $g = 2$ in the 0-Sphere Model: U(1) Fiber Bundle and Dual-Frame Phase Decomposition*, Zenodo (2026), [10.5281/zenodo.19227518](https://zenodo.org/record/19227518).
- [38] E. A. Coddington and N. Levinson, *Theory of Ordinary Differential Equations* (McGraw-Hill, New York, 1955).
- [39] J. A. Lissajous, *Mémoire sur l'étude optique des mouvements vibratoires*, *Ann. Chim. Phys.* (3) **51**, 147 (1857).
- [40] L. Onsager, *Statistical Hydrodynamics*, *Nuovo Cimento* **6**, Suppl. 2, 279 (1949).
- [41] R. P. Feynman, *Application of Quantum Mechanics to Liquid Helium*, in *Progress in Low Temperature Physics*, Vol. 1, edited by C. J. Gorter (North-Holland, Amsterdam, 1955), p. 17.
- [42] W. F. Vinen, *The Detection of Single Quanta of Circulation in Liquid Helium II*, *Proc. R. Soc. Lond. A* **260**, 218 (1961).

- [43] Bureau International des Poids et Mesures, *The International System of Units (SI)*, 9th ed. (BIPM, Sèvres, 2019).
- [44] E. Tiesinga, P. J. Mohr, D. B. Newell, and B. N. Taylor, *CODATA Recommended Values of the Fundamental Physical Constants: 2018*, Rev. Mod. Phys. **93**, 025010 (2021).
- [45] O. Klein, *Quantentheorie und fünfdimensionale Relativitätstheorie*, Z. Phys. **37**, 895 (1926).
- [46] W. Gordon, *Der Comptoneffekt nach der Schrödingerschen Theorie*, Z. Phys. **40**, 117 (1926).
- [47] S. Hanamura, *Coexistence of Positive- and Negative-Energy States in the Dirac Framework within the 0-Sphere Model*, Zenodo, [10.5281/zenodo.17760132](https://doi.org/10.5281/zenodo.17760132).
- [48] E. Schrödinger, *Über die kräftefreie Bewegung in der relativistischen Quantenmechanik*, Sitzungsber. Preuss. Akad. Wiss. Phys.-Math. Kl. **24**, 418 (1930).
- [49] B. Thaller, *The Dirac Equation* (Springer, Berlin, 1992).
- [50] L. L. Foldy and S. A. Wouthuysen, *On the Dirac Theory of Spin 1/2 Particles and Its Non-Relativistic Limit*, Phys. Rev. **78**, 29 (1950).
- [51] V. I. Arnold, *Mathematical Methods of Classical Mechanics*, 2nd ed. (Springer, New York, 1989).
- [52] F. Bloch, *Nuclear Induction*, Phys. Rev. **70**, 460 (1946).
- [53] H. Hopf, *Über die Abbildungen der dreidimensionalen Sphäre auf die Kugelfläche*, Math. Ann. **104**, 637 (1931).
- [54] T. Needham, *Visual Complex Analysis* (Oxford University Press, Oxford, 1997).
- [55] S. Hanamura, *Uniquely Distinguishing an Electron's Spin from Two Quantum States via Riemann Surface Guidance*, Zenodo (2019), [10.5281/zenodo.17759634](https://doi.org/10.5281/zenodo.17759634).
- [56] J. Stefan, *Über die Beziehung zwischen der Wärmestrahlung und der Temperatur*, Sitzungsber. Kais. Akad. Wiss. Wien, Math.-Naturwiss. Cl. **79**, 391 (1879).
- [57] L. Boltzmann, *Ableitung des Stefan'schen Gesetzes, betreffend die Abhängigkeit der Wärmestrahlung von der Temperatur aus der electromagnetischen Lichttheorie*, Ann. Phys. **258**, 291 (1884).
- [58] S. Hanamura, *The Square Root of the Hyperspherical Laplacian: A Geometric Foundation for Spin Two-Valuedness on S^3* , Zenodo (2026), [10.5281/zenodo.20820646](https://doi.org/10.5281/zenodo.20820646).
- [59] A. Lichnerowicz, *Spineurs harmoniques*, C. R. Acad. Sci. Paris **257**, 7 (1963).
- [60] G. C. Wick, *Properties of Bethe-Salpeter Wave Functions*, Phys. Rev. **96**, 1124 (1954).
- [61] L. H. Thomas, *The Motion of the Spinning Electron*, Nature **117**, 514 (1926).
- [62] L. H. Thomas, *The Kinematics of an Electron with an Axis*, Philos. Mag. (7) **3**, 1 (1927).
- [63] S. Hanamura, *Spin as a Real Vector from Internal Photon-Sphere Motion: Geometric Origin of $U(1)$ Gauge and $SU(2)$ Periodicity*, Zenodo (2025), [10.5281/zenodo.17765238](https://doi.org/10.5281/zenodo.17765238).
- [64] X. Fan, T. G. Myers, B. A. D. Sukra, and G. Gabrielse, *Measurement of the Electron Magnetic Moment*, Phys. Rev. Lett. **130**, 071801 (2023).
- [65] S. Hanamura, *Spin-Induced Inertial Resistance in Electrons: A Gyroscopic Interpretation Based on General Relativity*, Zenodo (2025), [10.5281/zenodo.17765121](https://doi.org/10.5281/zenodo.17765121).
- [66] S. Hanamura, *The Bridge Equation $\gamma = 1 + a$ from First Principles: Representation Duality of the Anomalous Moment and the Geometric Origin of the Root-Mean-Square Factor in the 0-Sphere Model*, Zenodo (2026), [10.5281/zenodo.20091680](https://doi.org/10.5281/zenodo.20091680).
- [67] J. Schwinger, *On Quantum-Electrodynamics and the Magnetic Moment of the Electron*, Phys. Rev. **73**, 416 (1948).
- [68] T. Aoyama, T. Kinoshita, and M. Nio, *Theory of the Anomalous Magnetic Moment of the Electron*, Atoms **7**, 28 (2019).
- [69] S.-Y. Lan, P.-C. Kuan, B. Estey, D. English, J. M. Brown, M. A. Hohensee, and H. Müller, *A Clock Directly Linking Time to a Particle's Mass*, Science **339**, 554 (2013).
- [70] D. N. Page and W. K. Wootters, *Evolution without Evolution: Dynamics Described by Stationary Observables*, Phys. Rev. D **27**, 2885 (1983).
- [71] C. Rovelli, *Quantum Gravity* (Cambridge University Press, Cambridge, 2004).
- [72] C. M. Will, *The Confrontation between General Relativity and Experiment*, Living Rev. Relativ. **17**, 4 (2014).
- [73] C. W. Chou, D. B. Hume, T. Rosenband, and D. J. Wineland, *Optical Clocks and Relativity*, Science **329**, 1630 (2010).
- [74] S. Hanamura, *Thermal Geodesics in the 0-Sphere Model: Extending General Relativity through Internal Thermodynamic Structure*, Zenodo (2025), [10.5281/zenodo.17765349](https://doi.org/10.5281/zenodo.17765349).
- [75] W. Gordon, *Zur Lichtfortpflanzung nach der Relativitätstheorie*, Ann. Phys. **377**, 421 (1923).
- [76] R. V. Pound and G. A. Rebka, *Apparent Weight of Photons*, Phys. Rev. Lett. **4**, 337 (1960).
- [77] S. Hanamura, *Gravitational Redshift of Internal Quantum Clocks: A Zitterbewegung-Based Model*, Zenodo (2025), [10.5281/zenodo.17765318](https://doi.org/10.5281/zenodo.17765318).
- [78] R. F. Marzke and J. A. Wheeler, *Gravitation as Geometry I: The Geometry of Space-Time and the Geometrodynamical Standard Meter*, in *Gravitation and Relativity*, edited by H.-Y. Chiu and W. F. Hoffmann (Benjamin, New York, 1964), p. 40.
- [79] F. W. Hehl and Yu. N. Obukhov, *Foundations of Classical Electrodynamics: Charge, Flux, and Metric* (Birkhäuser, Boston, 2003).
- [80] F. Kottler, *Maxwell'sche Gleichungen und Metrik*, Sitzungsber. Akad. Wiss. Wien, Math.-Naturwiss. Kl., Abt. IIa **131**, 119 (1922).
- [81] É. Cartan, *Sur les variétés à connexion affine et la théorie de la relativité généralisée (première partie)*, Ann. Sci. Éc. Norm. Supér. **40**, 325 (1923).
- [82] D. van Dantzig, *The Fundamental Equations of Electromagnetism, Independent of Metrical Geometry*, Proc. Camb. Philos. Soc. **30**, 421 (1934).
- [83] S. Hanamura, *Maxwell Electrodynamics as a Derived Low-Energy Limit of the 0-Sphere Line-Integral Ontology*, Zenodo (2026), [10.5281/zenodo.19701297](https://doi.org/10.5281/zenodo.19701297).
- [84] S. Hanamura, *From Clock Synchronization to Electromagnetism: A Realist Construction of $U(1)$ Gauge Theory*, Zenodo (2025), [10.5281/zenodo.17765136](https://doi.org/10.5281/zenodo.17765136).
- [85] H. Weyl, *Elektron und Gravitation. I*, Z. Phys. **56**, 330 (1929).
- [86] K. G. Wilson, *Confinement of Quarks*, Phys. Rev. D **10**, 2445 (1974).

- [87] T. T. Wu and C. N. Yang, *Concept of Nonintegrable Phase Factors and Global Formulation of Gauge Fields*, Phys. Rev. D **12**, 3845 (1975).
- [88] Y. Aharonov and D. Bohm, *Significance of Electromagnetic Potentials in the Quantum Theory*, Phys. Rev. **115**, 485 (1959).
- [89] A. Tonomura, N. Osakabe, T. Matsuda, T. Kawasaki, J. Endo, S. Yano, and H. Yamada, *Evidence for Aharonov–Bohm Effect with Magnetic Field Completely Shielded from Electron Wave*, Phys. Rev. Lett. **56**, 792 (1986).
- [90] S. Hanamura, *Geometric Structure of Spinorial Phase Accumulation along Thermal Geodesics*, Zenodo (2025), [10.5281/zenodo.18067760](https://zenodo.org/record/105281/18067760).
- [91] S. Hanamura, *From Curvature to Connection: Revisiting the Geometric Origin of Conservation Laws*, Zenodo (2026), [10.5281/zenodo.18819682](https://zenodo.org/record/105281/18819682).
- [92] S. Hanamura, *Line Integrals as Fundamental Observables in Physics: A Unified Principle Behind the Aharonov–Bohm Effect, Berry Phase, and Wilson Loops*, Zenodo (2026), [10.5281/zenodo.18203433](https://zenodo.org/record/105281/18203433).
- [93] S. Hanamura, *From Line Integral to Covariant Derivative: A Reader’s Map of the Bridge from the 0-Sphere Model to Riemannian Curvature*, Zenodo (2026), [10.5281/zenodo.20767589](https://zenodo.org/record/105281/20767589).
- [94] G. Breit, *An Interpretation of Dirac’s Theory of the Electron*, Proc. Natl. Acad. Sci. USA **14**, 553 (1928).
- [95] G. Bressi, G. Carugno, F. Della Valle, G. Galeazzi, G. Ruoso, and G. Sartori, *Testing the Neutrality of Matter by Acoustic Means in a Resonant Cavity*, Phys. Rev. A **83**, 052101 (2011).
- [96] S. Hanamura, *Binding Energy, Composition, and the Residual β_{ZB} : Two Tests of the Weak Equivalence Principle in the 0-Sphere Model, and a Restart of the Formalisation*, Zenodo (2026), in preparation.
- [97] S. Hanamura, *On Observer-Dependent Torsion, Zitterbewegung, and Phase Accumulation in the 0-Sphere Model*, Zenodo, [10.5281/zenodo.18896784](https://zenodo.org/record/105281/18896784).
- [98] C. R. Harris *et al.*, *Array Programming with NumPy*, Nature **585**, 357 (2020).
- [99] J. J. Monaghan, *Smoothed Particle Hydrodynamics*, Annu. Rev. Astron. Astrophys. **30**, 543 (1992).
- [100] J. S. Bell, *On the Problem of Hidden Variables in Quantum Mechanics*, Rev. Mod. Phys. **38**, 447 (1966).